

Notes on General Relativity and Cosmology

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These notes are an outgrowth of my lecture notes from my general relativity and cosmology course (PHYS 172) taught by Professor Rahulkumar Solanki at Harvey Mudd in Spring 2026. The following additional resources were used to compile these notes.

- *The Classical Theory of Fields* by Landau & Lifshitz. Prof. Solanki's favorite. He followed it quite closely in some sections of the course. Terse introduction to the subject from a legendary series of books.
- *General Relativity* by Wald. Graduate textbook; as Scott Hughes calls it, the "GR überbuch".
- *Wormholes, Warp Drives and Energy Conditions*, edited by Lobo.
- L. Heisenberg's *Review of $f(Q)$ gravity*. Paper that covers quite well the "geometrical trinity of gravity", which refers to three mathematically equivalent classical descriptions of Einstein's general relativity: standard GR (gravity encoded by curvature), teleparallel equivalent of general relativity (gravity encoded by torsion), and symmetric teleparallel equivalent of general relativity (gravity encoded by non-metricity). This is more of an interesting area of active research than part of a standard sequence.
- Baumann's notes on cosmology for Part III of Cambridge's mathematical tripos. Recommended to me by Professor Setter. (IN PROGRESS)

We use the standard Einstein convention (e.g. $a^i b_i \equiv \sum_{i=1}^3 a^i b_i = a^1 b_1 + a^2 b_2 + a^3 b_3$ for $i \in \{1, 2, 3\}$). In the GR part of these notes we use the mostly-plus metric signature and set $G = 1$, and in the cosmology part we use the mostly-minus signature and don't set $G \neq 1$. **Distinguishing GR and cosmology parts and naming the sections properly is something I'll do later.** I have occasionally suppressed explicit tensor-slot separation where the ordering is unambiguous. In such cases, superscripted indices are understood to precede subscripted indices in the slot ordering; for example, $R_{\rho\sigma}^{\mu\nu} := R^{\mu\nu}{}_{\rho\sigma}$.

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1 Principle of equivalence

- In Newtonian mechanics, the inertial mass is defined by $\vec{F} = m_i \vec{a}$ and the gravitational mass is defined by $\vec{F}_g = -m_g \vec{\nabla} \Phi$, where Φ is the gravitational field.
- The *weak equivalence principle* states that $m_i = m_g$, implying that the behavior of freely falling test particles does not depend on their mass. Consequently, the motion of freely falling particles is the same in a uniform gravitational field and a uniformly accelerated frame.
- The *Einstein equivalence principle* generalizes the statement to all local experiments. It tells us that there are no "gravitationally neutral" objects with respect to which we can measure g . Thus we instead define unaccelerated/inertial frames to be freely falling, which motivates us to describe gravity as not a force, but a property of spacetime.
- Since gravitational fields are inhomogeneous, global inertial frames don't exist.
- Consider an inertial frame K and a uniformly rotating frame of reference K' . In a plane perpendicular to the axis of rotation, the ratio of the circumference to the radius of a circle centered at the origin (through which the axis of rotation passes) in the inertial frame K is 2π . However, a meterstick at rest in K' experiences a Lorentz contraction in the azimuthal direction (viewed from inertial frame K) but not in radial direction since it is perpendicular to the relative velocity (once can consider instantaneous inertial frame of reference coinciding with K' at any instance). Therefore, the ratio of circumference to the radius of the same circle is greater than 2π . Thus, the space is non-Euclidean (or curved) in K' ; however, the spacetime is flat (i.e. Minkowski) because there exists a coordinate transformation through which one can return to inertial frame K . Therefore, from the principle of equivalence, gravitational fields have similar property in a small enough region of spacetime. However, "real" gravitational fields cannot be removed by coordinate transformations, so spacetime must be curved in their presence (which can be made locally flat using appropriate coordinate transformations). Thus, the mathematical theory of gravitational fields is the theory of curved spacetime.
- The propagation of light is also affected by gravitational fields since one cannot distinguish uniform acceleration from gravitational field locally using experiments consisting of light signals. In Newtonian theory, propagation of light has nothing to do with gravity. Furthermore, according to special relativity, the mass of a body is a measure of its energy-content; if the energy changes by l , the mass changes in the same sense by l/c^2 . For example, the mass of the hydrogen atom is less than the sum of masses of proton and electron due to negative binding energy. Therefore, according to the equivalence principle, the gravitational mass has to change in exactly the same manner. Essentially, any form of energy (whether mass or electromagnetic field) produces a gravitational field. Finally, the negative binding energy (gravitational or otherwise) makes the theory of general relativity highly nonlinear, since the gravitational field of the end product is not the vector sum of the field produced by constituents.

2 The geometry of spacetime

2.1 Curvilinear coordinates

- Consider the transformation of coordinates $x^\mu \rightarrow x'^\mu$. The basic transformation is that of the differential:

$$dx^i = \frac{\partial x^\mu}{\partial x'^\nu} dx'^\nu$$

- The derivatives of a scalar ϕ transform as

$$\frac{\partial \phi}{\partial x^i} = \frac{\partial \phi}{\partial x'^k} \frac{\partial x'^k}{\partial x^i}$$

- A *contravariant four-vector* is any aggregate of four quantities A^μ transforming like coordinate differentials under transformation of coordinates. The dual four-vectors are the *covariant four-vectors*. Together, these form the set of *rank-1 tensors* in four-dimensional spacetime¹. The transformation laws are

$$A^\mu = \frac{\partial x^\mu}{\partial x'^\nu} A'^\nu \quad (\text{contravariant vector})$$

$$A_\mu = \frac{\partial x'^\nu}{\partial x^\mu} A'_\nu \quad (\text{covariant vector})$$

- Rank-2 tensors* are aggregates of 16 quantities transforming like

$$A^{\mu\nu} = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x^\nu}{\partial x'^\sigma} A'^{\rho\sigma} \quad (\text{contravariant})$$

$$A_{\mu\nu} = \frac{\partial x'^\rho}{\partial x^\mu} \frac{\partial x'^\sigma}{\partial x^\nu} A'_{\rho\sigma} \quad (\text{covariant})$$

$$A^\mu{}_\nu = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x'^\sigma}{\partial x^\nu} A'^\rho{}_\sigma \quad (\text{mixed})$$

- More generally, a *tensor of type* (r, s) , or of rank $r + s$, is an object with r contravariant indices and s covariant indices whose components transform with one Jacobian factor for each index:

$$A^{\mu_1 \dots \mu_r}{}_{\nu_1 \dots \nu_s} = \frac{\partial x^{\mu_1}}{\partial x'^{\rho_1}} \dots \frac{\partial x^{\mu_r}}{\partial x'^{\rho_r}} \frac{\partial x'^{\sigma_1}}{\partial x^{\nu_1}} \dots \frac{\partial x'^{\sigma_s}}{\partial x^{\nu_s}} A'^{\rho_1 \dots \rho_r}{}_{\sigma_1 \dots \sigma_s}$$

Thus each upper index transforms like a contravariant vector index, and each lower index transforms like a covariant vector index.

- A *tensor density*² $\tilde{A}$ of weight w and type (r, s) is an object which transforms as

$$\tilde{A}^{\mu_1 \dots \mu_r}{}_{\nu_1 \dots \nu_s} = \left| \det \left(\frac{\partial x'}{\partial x} \right) \right|^w \frac{\partial x^{\mu_1}}{\partial x'^{\rho_1}} \dots \frac{\partial x^{\mu_r}}{\partial x'^{\rho_r}} \frac{\partial x'^{\sigma_1}}{\partial x^{\nu_1}} \dots \frac{\partial x'^{\sigma_s}}{\partial x^{\nu_s}} \tilde{A}'^{\rho_1 \dots \rho_r}{}_{\sigma_1 \dots \sigma_s}$$

- The *Kronecker delta* δ^μ_ν is a (rank-2) tensor which can "replace indices" like $A^\nu \delta^\mu_\nu = A^\mu$. Note:

$$\frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x^\mu}{\partial x'^\beta} = \delta^\alpha_\beta$$

$$\frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\sigma} = \delta^\rho_\sigma$$

¹From now on, when we speak of tensors, we assume that we are talking about tensors in four-dimensional spacetime, using the physicist's conventions.

²We will not use tensor densities until section 5.

- A spacetime is called *flat* if there exists a coordinate system in which the metric takes the form³ $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ everywhere. Such coordinates are called *galilean coordinates* or *global inertial coordinates*.
- Spacetimes coordinates x'^{μ} are said to be *curvilinear* if they are related to Galilean coordinates x^{μ} by a smooth, locally invertible transformation; generally, this transformation is nonlinear, so the coordinate lines need not be straight.
- An infinitesimal *spacetime interval* between two nearby events in curvilinear coordinates is

$$ds^2 = g_{\mu\nu} dx^{\mu} dx^{\nu}$$

where $g_{\mu\nu}$ are functions of the spacetime coordinates. $g_{\mu\nu}$ forms a symmetric covariant tensor called the *metric tensor*. The metric tensor raises/lowers indices: $A^{\mu} = g^{\mu\nu} A_{\nu}$, $A_{\mu} = g_{\mu\nu} A^{\nu}$. This also applies for tensors of higher rank; $A^{\mu}_{\nu} = g^{\mu\rho} A_{\rho\nu}$, $A^{\mu\nu} = g^{\mu\rho} g^{\nu\sigma} A_{\rho\sigma}$. In particular, $g_{\alpha\mu} g^{\beta\mu} = \delta^{\alpha}_{\beta}$.

- Let x'^{μ} be galilean coordinates and x^{μ} arbitrary curvilinear coordinates. Consider the Jacobian of the transformation from the galilean to curvilinear coordinates $J = \frac{\partial(x^0, x^1, x^2, x^3)}{\partial(x'^0, x'^1, x'^2, x'^3)}$. Then the determinant g of the metric tensor $g_{\mu\nu}$ in the system x^{μ} is given by $J = \frac{1}{\sqrt{-g}}$. So for example, in a galilean coordinate system the integral of a scalar with respect to the four-volume element $d\Omega' = dx'^0 dx'^1 dx'^2 dx'^3$ is also a scalar, so on transforming to curvilinear coordinates, the element of integration goes over into $d\Omega' \rightarrow \sqrt{-g} d\Omega$. Thus, in curvilinear coordinates, when integrating over a four-volume, the quantity $\sqrt{-g} d\Omega$ behaves like an invariant.
- Since g contains the square of a Jacobian determinant, it does not transform as an ordinary scalar; that is, $\sqrt{-g} = \left| \det \frac{\partial x'}{\partial x} \right| \sqrt{-g'}$. This lets us write a tensor density of weight w as $\tilde{A}^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} = (\sqrt{-g})^w A^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q}$ where $A^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q}$ is a tensor.
-
- From now on, we denote by τ the *proper time* (i.e. the time experienced in an object's rest frame). For two infinitesimally separated events happening at the same point in space, and denoting by $d\tau$ the (proper) time interval between the two events, we have $ds^2 = -c^2 d\tau^2$, from which $d\tau = \frac{1}{c} \sqrt{-g_{00}} dx^0$. So, the (proper) time between any two events occurring at the same point in space is

$$\tau = \frac{1}{c} \int \sqrt{-g_{00}} dx^0$$

2.2 Covariant differentiation and Christoffel symbols

In galilean coordinates, the differential dA^{μ} of a vector A^{μ} forms a vector, and the derivatives $\frac{\partial A^{\mu}}{\partial x^{\nu}}$ form a tensor. In general curvilinear coordinates, neither quantity is tensorial, due to the fact that dA^{μ} is the difference of vectors located at different, infinitesimally separated points of space. In order to obtain a differential of a vector that behaves like a vector. It is necessary that the two vectors to be subtracted from each other be located at the same point in space; that is, we translate one of the vectors to the point where the second is, after which we determine the

³This is the so called "east coast convention" used in GR. In quantum field theory, and in many treatments of GR, one uses the metric tensor $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

difference of two vectors which we now refer to one and the same point in space. We also need the operation of translation to be defined so that in galilean coordinates the difference coincides with dA^μ . Under parallel translation of a vector, its components in galilean don't change, but in curvilinear coordinates they generally do (the canonical example is a vector on the equator of a sphere, parallel transported to one of the poles, then back down to the equator in a direction perpendicular to the first displacement, and finally back to the original point on the equator; the final vector comes back rotated by 90 degrees). This ends up being precisely how we do calculus in curvilinear coordinates.

- The change in a vector under a *parallel displacement* is given by

$$DA^\mu = \left(\frac{\partial A^\mu}{\partial x^\rho} + \Gamma_{\nu\rho}^\mu A^\nu \right) dx^\rho$$

$$DA_\mu = \left(\frac{\partial A_\mu}{\partial x^\rho} - \Gamma_{\mu\rho}^\nu A_\nu \right) dx^\rho$$

where the quantities $\Gamma_{\nu\rho}^\mu$ are (local) coefficients called *Christoffel symbols*. The expressions in the parentheses, called the *covariant derivatives* of vectors A^μ , A_μ , denoted $\nabla_\rho A^\mu$ and $\nabla_\rho A_\mu$ respectively, give the desired generalization of the concept of a derivative to curvilinear coordinates.

- One can similarly determine the covariant derivative of a tensor of arbitrary rank; to obtain the covariant derivative of a tensor $A^{\dots}$ with respect to x^ρ , we add to the ordinary derivative $\partial A^{\dots}/\partial x^\rho$ for each covariant index ($A_{\mu\dots}$) a term $-\Gamma_{\mu\rho}^\nu A_{\nu\dots}$, and for each contravariant index ($A^{\dots\mu}$) a term $+\Gamma_{\nu\rho}^\mu A^{\dots\nu}$.
- In a Galilean coordinate system, the Christoffel symbols vanish, and covariant differentiation reduces to ordinary differentiation.
- In terms of the metric tensor,

$$\Gamma_{\nu\rho}^\mu = \frac{g^{\mu\sigma}}{2} (\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho}) = \Gamma_{\rho\nu}^\mu$$

- The Christoffel symbols are not tensors: they behave like a tensor only under linear transformations. Generally, they transform as

$$\Gamma_{\mu\lambda}^{\nu} = \frac{\partial x'^\nu}{\partial x^\alpha} \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\lambda} \Gamma_{\rho\sigma}^\alpha - \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\lambda} \frac{\partial^2 x'^\nu}{\partial x^\alpha \partial x^\beta}$$

- The covariant derivative as defined above indeed follows the conditions one requires of a derivative operator: linearity, the Leibnitz (product) rule, commutativity with contraction, consistency with the notion of tangent vectors as directional derivatives on scalar fields, and finally, the "torsion-free" condition $\nabla_\mu \nabla_\nu f = \nabla_\nu \nabla_\mu f$ for scalar f . While this last condition is sometimes dropped, in GR the derivative operator is assumed to be torsion-free.
- The metric tensor acts as a constant under covariant differentiation, that is, $\nabla_\rho g_{\mu\nu} = 0$; in fact, for any given metric, the derivative operator ∇_ρ satisfying this condition exists and is unique.
- If in a covariant derivative we raise the index signifying the differentiation, we obtain the so-called *contravariant derivative*; thus, $\nabla^\nu A_\mu = g^{\nu\rho} \nabla_\rho A_\mu$, $\nabla^\nu A^\mu = g^{\nu\rho} \nabla_\rho A^\mu$.

- It is always possible to choose a coordinate system in which all the $\Gamma_{\nu\rho}^\mu$ become zero at a previously assigned point. Such a system is said to be *locally inertial*. In fact, let a given point be chosen as the origin of coordinates, and let the values of $\Gamma_{\nu\rho}^\mu$ at it be initially (in the coordinates x^μ) equal to $(\Gamma_{\nu\rho}^\mu)_0$. In the neighborhood of this point, make the transformation $x'^\mu = x^\mu + \frac{1}{2}(\Gamma_{\nu\rho}^\mu)_0 x^\nu x^\rho$. Then all the $\Gamma_{\nu\rho}^\mu$ become zero. Such a transformation has $\left(\frac{\partial x'^\mu}{\partial x^\nu}\right)_0 = \delta_\nu^\mu$, so it does not change the value of any tensor at the given point, so we can make the Christoffel symbols vanish at the same time as we bring the $g_{\mu\nu}$ to Cartesian form.
- A vector A^μ given at each point on some curve C with a tangent B^μ is said to be *parallelly transported* as one moves along the curve if $B^\mu \nabla_\mu A^\nu = 0$ along the curve. More generally, one can define the parallel transport of a tensor of arbitrary rank by $B^\mu \nabla_\mu A^{\rho_1 \dots \rho_r}_{\sigma_1 \dots \sigma_s}$. Choosing a coordinate basis and parametrizing C by t , $\frac{dA^\nu}{dt} + B^\mu \Gamma_{\mu\lambda}^\nu A^\lambda = 0$.

2.3 Curvature

Euclid's fifth postulate states that if two lines are drawn which intersect a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough. Two-dimensional space of constant negative curvature satisfies all but this postulate of Euclid. There are two approaches to studying the curvature of an object: the *extrinsic approach*, which is to analyze the object by embedding it in a higher-dimensional space, or to analyze it *intrinsically* by measuring entirely from within, without referencing any higher-dimensional surrounding space. General relativity studies curved spacetime intrinsically.

- Let $\delta A_\mu = \Gamma_{\mu\nu}^\gamma A_\gamma dx^\nu$ be the change in DA_μ under the parallel transport from $x^\mu \rightarrow x^\mu + dx^\mu$, i.e. $DA_\mu = dA_\mu + \delta A_\mu$. This δA_μ is path-dependent due to the dx^ν term, and therefore so is the covariant difference. The change in an arbitrary vector field after parallel displacement around any infinitesimal closed contour gives us necessary and sufficient conditions for whether or not spacetime is curved. Write this change $\Delta A_\mu = \oint \delta A_\mu = \oint \Gamma_{\mu\nu}^\lambda A_\lambda dx^\nu$. Then, with the first-order accuracy sufficient for the transformation, $\Delta A_\mu = \frac{1}{2} R_{\mu\nu\sigma}^\lambda A_\lambda \Delta f^{\nu\sigma}$, where $\Delta f^{\nu\sigma}$ is the infinitesimal value of the area enclosed by the contour, and $R_{\mu\nu\sigma}^\lambda$ is the curvature tensor defined below.
- Define the *curvature tensor* (or *Riemann tensor*)

$$R_{\mu\nu\alpha\beta} A^\nu = (\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) A_\mu$$

Equivalently,

$$\begin{aligned} R_{\nu\rho\sigma}^\mu &= \partial_\rho \Gamma_{\nu\sigma}^\mu - \partial_\sigma \Gamma_{\nu\rho}^\mu + \Gamma_{\alpha\rho}^\mu \Gamma_{\nu\sigma}^\alpha - \Gamma_{\alpha\sigma}^\mu \Gamma_{\nu\rho}^\alpha \\ R_{\mu\nu\rho\sigma} &= \frac{1}{2} (\partial_\nu \partial_\rho g_{\mu\sigma} + \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\nu \partial_\sigma g_{\mu\rho} - \partial_\mu \partial_\rho g_{\nu\sigma}) + g_{\alpha\beta} (\Gamma_{\nu\rho}^\alpha \Gamma_{\mu\sigma}^\beta - \Gamma_{\nu\sigma}^\alpha \Gamma_{\mu\rho}^\beta) \end{aligned}$$

This tensor vanishes if and only if the curvature is zero. By way of coordinate transformations, we can make $\Gamma_{\beta\sigma}^\alpha = 0$ at a single point, but $\partial_\beta \Gamma_{\beta\gamma}^\alpha \neq 0$ if the spacetime is curved.

- Properties of the Riemann tensor include:

$$\begin{aligned} R_{\alpha\beta\gamma\delta} &= -R_{\beta\alpha\gamma\delta} = R_{\beta\alpha\delta\gamma} = R_{\gamma\delta\alpha\beta} g^{\mu\nu} R_{\mu\nu\rho\sigma} = R_{\mu\rho\sigma}^\mu = 0 \\ R_{\alpha\beta\gamma\delta} + R_{\beta\gamma\alpha\delta} + R_{\gamma\alpha\beta\delta} &= 0 \\ \nabla_\alpha R_{\beta\gamma\delta\sigma} + \nabla_\beta R_{\gamma\alpha\delta\sigma} + \nabla_\gamma R_{\alpha\beta\delta\sigma} &= 0 \end{aligned}$$

- The trace of the curvature (or Riemann) tensor is captured by the *Ricci tensor* $R_{\nu\sigma} = g^{\mu\rho} R_{\mu\nu\rho\sigma} = R_{\nu\mu\sigma}^{\mu}$ and tracefree part is captured by the *Weyl tensor* $C_{\alpha\beta\gamma\delta} = R_{\alpha\beta\gamma\delta} - \frac{1}{2}R_{\alpha\gamma}g_{\beta\delta} + \frac{1}{2}R_{\alpha\delta}g_{\beta\gamma} + \frac{1}{2}R_{\beta\gamma}g_{\alpha\delta} - \frac{1}{2}R_{\beta\delta}g_{\alpha\gamma} + \frac{R}{6}(g_{\alpha\gamma}g_{\beta\delta} - g_{\alpha\delta}g_{\beta\gamma})$. In some notations, one writes $C_{\rho\sigma}^{\mu\nu} = R_{\rho\sigma}^{\mu\nu} - 2\delta_{[\rho}^{\mu}R_{\sigma]}^{\nu]} + \frac{R}{3}\delta_{[\rho}^{\mu}\delta_{\sigma]}^{\nu]}$, where the square bracket around indices represents *antisymmetrization* $A_{[\alpha\beta]} = \frac{1}{2}(A_{\alpha\beta} - A_{\beta\alpha})$. There exists also the *symmetrization* $A_{(\alpha\beta)} = \frac{1}{2}(A_{\alpha\beta} + A_{\beta\alpha})$.
- The *Ricci scalar* is $g^{\nu\sigma} R_{\nu\sigma} = R$.

2.4 Geodesics

- The equations of motion for a free motion of a particle in special relativity are $\frac{du^\mu}{d\tau} = 0$ or $du^\mu = 0$, where $u^\mu = \frac{dx^\mu}{d\tau}$ is the *four-velocity*. It follows by definition that $u^\mu u_\mu = -1$.
- We postulate that the motion of freely falling massive particles (i.e. particles traveling with speed less than c under the influence of only gravity) is given by the path of maximum proper time, called a *timelike geodesic*. Physically, this postulate makes sense because such paths are locally straight through spacetime, just like segments of minimum length are straight lines in Euclidean space. A physical approach to finding this motion would be to set $a^\mu = \frac{Du^\mu}{D\tau} = 0$. Equivalently, such particles satisfy the equations of motion

$$\frac{du^\mu}{d\tau} + \Gamma_{\nu\rho}^\mu u^\nu u^\rho = \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0$$

We call this the *timelike geodesic equation*.

- Alternatively, we can define a timelike geodesic to be a timelike curve $x^\mu = x^\mu(\sigma)$ joining two events A and B , where σ is some parameter, satisfying

$$\delta S = \delta \left(\int_A^B \sqrt{-g_{\mu\nu}} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} d\sigma \right) = 0$$

among all the timelike curves in the vicinity joining A and B . Recall that the tangent to such a timelike curve is $\frac{dx^\mu}{d\sigma}$. Curves satisfying the above then follow the Euler-Lagrange equations $\frac{\partial \mathcal{L}}{\partial x^\alpha} - \frac{d}{d\sigma} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) = 0$ where $L = \sqrt{-g_{\mu\nu}} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}$; then the first definition is recovered.

- A spacelike path cannot be travelled by massive or massless particles since the events lying on these paths are non-causal. However, spacelike geodesics can be useful in defining spatial coordinates, for example, in the case of Fermi normal coordinates. If σ is a parameter for a spacelike curve $x^\mu = x^\mu(\sigma)$, then $g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} > 0$ and one can proceed in a similar manner as the previous parts to find a geodesic equation with proper length s (instead of τ):

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{ds} \frac{dx^\rho}{ds} = 0$$

- For a null curve, $ds^2 = 0$, so it does not make sense to apply the variational principle since all null paths will give total spacetime interval between fixed events A, B to equal zero. If $x^\mu = x^\mu(\lambda)$ is a null curve, then one generally chooses λ such that the wave four-vector is $k^\mu = \frac{dx^\mu}{d\lambda}$. Since we consider geodesics as generalizations of straight lines, $Dk^\mu = 0$ is the geodesic equation. We have thus obtained the *lightlike geodesic equation*

$$\frac{dk^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu k^\nu k^\sigma = 0 = \frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda}$$

- It is very important to understand the difference between a null path and a null geodesic; for a null path, $k^\mu k_\mu = 0$, and null geodesics satisfy the geodesic equation in addition to the null path condition.

3 Einstein's general relativity

3.1 The energy-momentum tensor

- The *(energy-)momentum four-vector* p^μ of a particle of mass m is defined by $p^\mu = mu^\mu$. The *energy* of a particle as measured by an observer present at the site of the particle whose 4-velocity is v^μ is defined by $E = -p_\mu v^\mu$. Thus, in special relativity, energy is recognized to be the time component of p^μ . For a particle at rest with respect to the observer (i.e. $v^\mu = u^\mu$), this reduces to the familiar $E = mc^2$. From now on, we use natural units, setting $c = 1$.
- We find that $p^\mu = mu^\mu$, $p^\mu p_\mu = -m^2$.
- Continuous matter distributions in special relativity are described by a symmetric *(stress-)energy-momentum* tensor.
- Consider the Lorentz transformation

$$\begin{aligned} x'^0 &= \frac{x^0 - ux^1}{\sqrt{1 - u^2}} & x'^1 &= \frac{x^1 - ux^0}{\sqrt{1 - u^2}} \\ x'^2 &= x^2 & x'^3 &= x^3 \end{aligned}$$

moving from a frame S to a frame S' moving at a velocity u in the $+x$ direction relative to S . The line element $d\ell^2 = -dt^2 + dx^2 + dy^2 + dz^2$ is invariant under such transformations.

- Under such transformations, a four-vector A^μ transforms as

$$\begin{aligned} A'^0 &= \frac{A^0 - uA^1}{\sqrt{1 - u^2}} & A'^1 &= \frac{A^1 - uA^0}{\sqrt{1 - u^2}} \\ A'^2 &= A^2 & A'^3 &= A^3 \end{aligned}$$

- Unlike mass, density is not an invariant, so it transforms under Lorentz transformations. For the treatment of charge density (which is equivalent to that of number density), consider the charge enclosed by some volume V in S . When we Lorentz-transform, the charge in the corresponding volume V' in S' is equal to the charge enclosed by V . Thus $\rho dV = \rho' dV' = \rho_0 dV_0$, where the last quantity refers to the proper frame, i.e. where the charges are at rest. Let v, v' denote the velocities of S, S' w.r.t. the proper frame, respectively. Then:

$$\begin{aligned} dV &= dV_0 \sqrt{1 - v^2} & \rho &= \frac{\rho_0}{\sqrt{1 - v^2}} \\ dV' &= dV_0 \sqrt{1 - v'^2} & \rho' &= \frac{\rho_0}{\sqrt{1 - v'^2}} \end{aligned}$$

yielding $\rho' = \rho \sqrt{\frac{1 - v^2}{1 - v'^2}}$. Recalling velocity transformations $v' = \frac{v - u}{1 - vu}$,

$$\rho' = \frac{\rho(1 - uv)}{\sqrt{1 - u^2}} = \frac{\rho - u(\rho v)}{\sqrt{1 - v^2}}$$

- $\vec{J} = \rho\vec{v}$ is the *charge flux* or *current density*, i.e. the charges per unit area per unit time.
- Now consider a case in which the charges are in S , moving in a direction normal to some fixed cross sectional area A at a velocity v . In time dt , the amount of charges passing through A is given by $\rho A v dt$, so the charge flux is ρv . Now, $\rho' v' = \frac{\rho(1-uv)}{\sqrt{1-u^2}} \left(\frac{v-u}{1-uv} \right) = \frac{\rho(v-u)}{\sqrt{1-u^2}}$.
- We construct the *four-current density*

$$J^\mu = (\rho, \vec{J}) = (\rho, \rho\vec{v}) = \rho_0 \left(\frac{1}{\sqrt{1-v^2}}, \frac{\vec{v}}{\sqrt{1-v^2}} \right) = \rho_0 u^\mu$$

- A *global conservation* law for some quantity (e.g. for charge or energy) tells us that the integral of that quantity's density over all volume is constant. Then *local conservation* needs to hold for causality; this yields a *continuity* condition that is, for some arbitrary volume V enclosed by a surface S , local conservation requires that the rate of change of total charges inside V equal the number of charges leaving or entering surface S per unit time. Mathematically,

$$\begin{aligned} -\frac{\partial}{\partial t} \iiint_V \rho dV &= \oint_S \vec{J} \cdot d\vec{S} \\ \Rightarrow \iiint_V -\frac{\partial \rho}{\partial t} dV &= \iiint_V (\nabla \cdot \vec{J}) dV \\ \Rightarrow \frac{\partial S}{\partial t} + \nabla \cdot \vec{J} &= 0 \end{aligned}$$

In terms of four-vectors, we can write $\frac{\partial J^\mu}{\partial x^\mu} = 0$. The covariant version is simply $\nabla_\mu J^\mu = 0$.

- For energy density, we have the same setup. Because energy is a component of the four-momentum, which is a four-vector, it transforms under a Lorentz transform accordingly:

$$E' = \frac{E - up_x}{\sqrt{1-u^2}}, \quad p'_x = \frac{p_x - uE}{\sqrt{1-u^2}}$$

- Energy flux in the $+x$ direction is the amount of energy passing through unit area in yz plane in unit time. Thus energy flux equals energy density $\frac{\rho_0}{\sqrt{1-v^2}}$ times velocity; this is just then momentum density. This gives us the *energy-momentum tensor for pressureless dust*:

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

- For a particle, a rank 1 four-momentum tells us how energy and momentum transform from one frame to another; for a continuous distribution of matter, a rank 2 energy-momentum tensor tells us how energy and momentum densities transform from one frame to another.
- T^{00} is the energy density, T^{i0} is the (3-)momentum density, T^{0i} is the energy flux, and T^{ik} is the stress tensor or momentum flux. $T^{\mu\nu} = T^{\nu\mu}$ due to energy flux equaling momentum density and the stress tensor being symmetric (due to conservation of angular momentum).
- In terms of $T^{\mu\nu}$, conservation of energy is $\nabla_\mu T^{0\mu} = 0$, and conservation of momentum (in the i th direction) is $\nabla_\mu T^{i\mu} = 0$. Thus, in general, $\nabla_\mu T^{\mu\nu} = 0$.

- Instead of pressureless dust, consider a perfect fluid with isotropic pressure. In the rest/proper/comoving frame of the fluid, there is only isotropic pressure P and no dissipation or energy flux. Therefore, for a perfect fluid in the rest frame,

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

In general,

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu}$$

The electromagnetic field behaves somewhat like a pressurized fluid; it turns out that for an electromagnetic field $T^{\mu\nu} = F^{\mu\alpha}F^\nu_\alpha - \frac{g^{\mu\nu}}{4}F_{\alpha\beta}F^{\alpha\beta}$ where $F^{\mu\nu}$ is the electromagnetic tensor.

- The most general form of the energy-momentum tensor is

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + q^\mu u^\nu + q^\nu u^\mu + Pg^{\mu\nu} + \pi^{\mu\nu}$$

where q^μ is energy flux and $\pi^{\mu\nu}$ is (traceless) anisotropic pressure.

3.2 Einstein's field equations

- Matter tells spacetime how to curve and spacetime tells matter how to move. The curvature of spacetime should then correspond to the energy-momentum of matter/fluid. But the latter is (4-)divergence-free, i.e. $\nabla_\mu T^{\mu\nu} = 0$, whereas the rank-2 tensor we use to describe curvature—the Ricci tensor—is not divergence-free. Using the Bianchi identity, one defines a divergence-free tensor, called the *Einstein tensor*:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{R}{2}g_{\mu\nu}$$

- *Einstein's field equations* are

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

- The formal classical field theory approach to derive this is to use the *Einstein-Hilbert* action

$$S = S_{\text{gravity}} + S_{\text{matter}} = \int R\sqrt{-g}d\Omega + S_{\text{matter}}$$

where $g = \det(g_{\mu\nu})$ and $d\Omega$ is a four-volume element and $\delta S_{\text{matter}} = \int T_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}d\Omega$. This can be compared to classical electrodynamics, where

$$S = S_{\text{field}} + S_{\text{interactions}} + S_{\text{matter}} = \int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}d\Omega + \int A_\mu J^\mu\sqrt{-g}d\Omega + \int \mu\sqrt{-g}d\Omega$$

where μ is mass density. The equation $\delta S_{\text{interactions}} + \delta S_{\text{matter}} = 0$ gives the Lorentz force $F^{\alpha\beta}J_\beta = \mu a^\alpha$, and $\delta S_{\text{field}} + \delta S_{\text{interactions}} = 0$ gives the field equation $\nabla_\mu F^{\mu\nu} = -J^\nu$.

- Rank 2 tensors have 16 components. Symmetric rank 2 tensors thus have 10 independent components. Conservation of energy $\nabla_\mu T^{\mu\nu} = 0$ means there are 4 constraints or 4 degrees of freedom that can be gauge fixed. Thus, there are 6 "independent" field equations.

3.3 Schwarzschild

- The first *exact* solution of Einstein's field equation was found by Schwarzschild. He gave two solutions, interior and exterior, where the former refers to the gravitational field inside a spherically symmetric constant density star, whereas the latter refers to the gravitational field outside any spherically symmetric star, that is, a vacuum solution.
- In vacuum, $T_{\mu\nu} = 0 = R_{\mu\nu}$, thus $R = 0$.
- A solution is called *asymptotically flat* if, infinitely far away from the spherical body producing the gravitational field, the spacetime becomes flat/Minkowski.
- A solution is called *static* if spacetime does not change over time and is also irrotational. This is a spatial case of *stationary spacetime*, in which the geometry does not change in time but can rotate.
- *Birkhoff's theorem* states that any spherically symmetric solution of the vacuum field equations (e.g. the Schwarzschild exterior solution, which is unique) is static and asymptotically flat.
- The solution being static means that spherically symmetric gravitational waves don't exist, even though matter in spherical stars is allowed to move/oscillate radially.
- We consider the static solution. A static solution has a line element of the form

$$ds^2 = g_{tt}dt^2 + g_{ik}dx^i dx^k$$

where $\partial_t g_{\mu\nu} = 0$.

- The most general spherically symmetric static line element is

$$ds^2 = -e^{2v(r)}dt^2 + e^{2\lambda(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

The Einstein tensor is then

$$G_t^t = -\frac{e^{2\lambda}}{r^2}(2r\lambda' + e^{2\lambda} - 1) \quad G_r^r = -\frac{e^{2\lambda}}{r^2}(e^{2\lambda} - 1 - 2rv')$$

$G_\theta^\theta = G_\phi^\phi \neq 0$ but we don't need it to find $\lambda(r), v(r)$. Now, $G_t^t = 0 \implies \frac{d}{dr}(re^{-2\lambda}) = 1 \implies e^{-2\lambda} = (1 - \frac{r_s}{r})$, where r is an integration constant; furthermore, $G_r^r = 0$ and $e^{2v} = (1 - \frac{r_s}{r})$. We take $r_s = 2M$ for reasons which will be clear in the next section. Basically, in the weak field limit, M is mass or newtonian mass.

- Thus the exterior Schwarzschild solution reduces to

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{\left(1 - \frac{2M}{r}\right)} + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

If one inserts G and c back then $|g_{tt}| = 1 - \frac{2GM}{rc^2}$.

3.4 The Newtonian limit

- Consider a spherically symmetric gravitational field produced by a star in Newtonian gravity. The equation of motion of a test particle in a radial ingoing free fall is given by $\vec{a} = -\nabla\Phi \implies \frac{d^2r}{dt^2} = -\frac{M}{r^2}$ where we take $G = 1$.
- The same notion in GR is the geodesic equation with $\frac{d\theta}{d\tau} = \frac{d\phi}{d\tau} = 0$. We consider equatorial plane without loss of generality i.e. $\theta = \pi/2$. One computes the Christoffel symbols

$$\begin{aligned}\Gamma_{tt}^r &= \frac{M}{r^2} \left(1 - \frac{2M}{r}\right) & \Gamma_{rt}^r &= \frac{-M}{r^2} \left(1 - \frac{2M}{r}\right) = -\Gamma_{tr}^r \\ \Gamma_{r\theta}^\theta &= \frac{1}{r} = \Gamma_{r\phi}^\phi & \Gamma_{\theta\theta}^r &= -r \left(1 - \frac{2M}{r}\right) \\ \Gamma_{\phi\phi}^r &= -r \sin^2 \theta \left(1 - \frac{2M}{r}\right) & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ & & \Gamma_{\theta\phi}^\phi &= \cot \theta\end{aligned}$$

- To compare Newtonian and general relativistic trajectories, far from the body producing gravitational field, coordinate length and time become proper length and time due to asymptotic flatness; in otherwords, we need to compare $\frac{d^2r}{dt^2}$ for GR using geodesic equation where r is quite large. Using

$$\frac{d^2r}{d\tau^2} = \frac{d^2r}{dt^2} \dot{t}^2 + \frac{dr}{dt} \ddot{t}$$

the geodesic equations for r and t become

$$\begin{aligned}\ddot{r} + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{tt}^r \dot{t}^2 &= 0 \\ \ddot{t} + \Gamma_{tr}^t \dot{r} \dot{t} &= 0\end{aligned}$$

Plugging and simplifying and reinserting G, c ,

$$\frac{d^2r}{dt^2} = -\frac{GM}{r^2} + \frac{2G^2M^2}{r^3c^2}$$

so in the newtonian limit $t \rightarrow \tau$ and $c \rightarrow \infty$ we recover regular equation of motion for Newtonian gravity. Thus, the parameter M in the Schwarzschild solution is mass in the Newtonian limit.

3.5 Newtonian orbits

In the following discussion, we work in polar coordinates. By conservation of energy, for a Newtonian orbit $E_n = K + V = \frac{1}{2}mv^2 - \frac{GMm}{r}$ is conserved. The equation of motion in r , with $\dot{\theta}$ expressed in terms of conserved angular momentum $l = mr^2\dot{\theta}$, involves only r and its derivatives. It is the same equation as would be obtained for a fictitious one-dimensional problem in which a particle of mass m is subject to a force $f' = f + \frac{l^2}{mr^3}$. The significance of the additional term is clear if it is written as $mr\dot{\theta}^2 = \frac{mv_\theta^2}{r}$, which is the familiar centrifugal force. Then the equation of motion of the particle in r is that of a one-dimensional problem with a fictitious potential $V' = V + \frac{1}{2} \frac{l^2}{mr^2}$. So the effective potential energy is $V = -\frac{GMm}{r} + \frac{l^2}{2mr^2}$; plotting this against r gives a potential well. At the bottom of this potential well are circular orbits. At energies where a horizontal line intersects the plotted curve at 2 points, we have elliptic orbits. If it intersects at only one point, we have parabolic/hyperbolic orbits; unless they are directly headed towards each other, the masses pass each other, never to meet again.

3.6 Measurements

- The energy of a particle measured by the observer moving with four-velocity u^μ is $E = -p_\mu u^\mu$.
- For massless particles, since our wave four-vector k^μ is related to p^μ by some constant, we can replace $p^\mu \rightarrow k^\mu$ and $E \rightarrow \omega$, yielding $\omega_0 = -k_\mu u^\mu$.

3.7 Energy conditions

- A *congruence* in an open subset of spacetime is a family of curves such that through each point in that subset there passes precisely one curve in this family. The tangents to a congruence yield a vector field on that subset, and every continuous vector field generates a congruence of curves. The congruence is smooth if the corresponding vector field is smooth.
- Consider a smooth congruence of timelike geodesics, which we parametrize by proper time τ , so that the vector field of tangents ξ^μ is normalized to unit length $\xi^\mu \xi_\mu = -1$. Then the tensor field $B_{\mu\nu}$ defined by $B_{\mu\nu} = \nabla_\nu \xi_\mu$ will satisfy $B_{\mu\nu} \xi^\mu = B_{\mu\nu} \xi^\nu = 0$. The physical significance of this is that an observer on a geodesic in the congruence would find the nearby geodesics surrounding him to be stretched and rotated by the linear map B_ν^μ . Define the "spatial metric" $h_{\mu\nu} = g_{\mu\nu} + \xi_\mu \xi_\nu$; h_ν^μ is the projection operator onto the subspace of the tangent space perpendicular to ξ^μ . Define the *expansion* θ , *shear* $\sigma_{\mu\nu}$, and *twist* $\omega_{\mu\nu}$ of the congruence by

$$\theta = B^{\mu\nu} h_{\mu\nu}, \quad \sigma_{\mu\nu} = B_{(\mu\nu)} - \frac{1}{3} \theta h_{\mu\nu}, \quad \omega_{\mu\nu} = B_{[\mu\nu]}$$

By Frobenius' theorem, the congruence is (locally) hypersurface orthogonal iff $\omega_{\mu\nu} = 0$.

- We have

$$\xi^\rho \nabla_\rho B_{\mu\nu} = -B_\nu^\rho B_{\mu\rho} + R_{\rho\nu\mu}{}^\sigma \xi^\rho \xi_\sigma$$

Taking the trace, we obtain *Raychaudhuri's equation*

$$\xi^\rho \nabla_\rho \theta = \frac{d\theta}{d\tau} = -\frac{1}{3} \theta^2 - \sigma_{\mu\nu} \sigma^{\mu\nu} + \omega_{\mu\nu} \omega^{\mu\nu} - R_{\rho\lambda} \xi^\rho \xi^\lambda$$

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3.8 Schwarzschild geodesics

- In the Schwarzschild metric, $L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} = \sqrt{\left| -f(r) \dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2 \dot{\phi}^2 \right|}$ where $\dot{t} = \frac{dt}{d\lambda}$ and $f(r) = \left(1 - \frac{2M}{r}\right)$.
- One can take a shortcut by using the Lagrangian $\mathcal{L} = -f(r) \dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2 \dot{\phi}^2$ where we parametrize directly using proper time i.e. $\dot{t} = \frac{dt}{d\tau}$. Take $\mathcal{L} = \epsilon$ where ϵ is a constant of motion, 0 for null geodesics and -1 for timelike geodesics, and the geodesic equation is $\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right)$.
- Generalized momenta are defined similarly to in the Lagrangian approach. For timelike geodesics, $\frac{p_\mu}{m} = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \Big|_{L=1, \tau=\lambda}$ where m is the mass of the particle.
- In general, we normalize four-velocity so $L^2 = -1$ and $L = 1$.

- Sind $\frac{\partial \mathcal{L}}{\partial t} = 0$, energy per unit rest mass is conserved, that is, $\frac{p_t}{m} = -E = -f(r)\frac{dt}{d\tau} = -f(r)\dot{t} = \text{constant}$.
- Similarly, since $\frac{\partial \mathcal{L}}{\partial \phi} = 0$, momentum per unit mass is conserved. That is, $\frac{p_\phi}{m} = J = r^2\dot{\phi} = \text{constant}$.
- So the conserved quantities are $\mathcal{L}$, E , and J .
- From the constants of motion,

$$\frac{E^2 - 1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 - \frac{M}{2} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3}$$

where effective potential

$$V_{eff} = -\frac{M}{r} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3} = -\left(\frac{r}{M}\right)^{-1} + \frac{1}{2} \left(\frac{J/M}{r/M}\right)^2 - \frac{(J/m)^2}{(r/M)^3}$$

To compare with Newtonian orbits let's reinstate c, G and write E as the energy per unit mass $\frac{E_n + mc^2}{mc^2}$ and J as the angular momentum per unit mass J_n/M . To take the Newtonian limit $c \rightarrow \infty$ we recover our usual Newtonian orbit expression ($\tau \rightarrow t$)

$$E_n = \frac{m}{2} \left(\frac{dr}{dt} \right)^2 + \frac{J_n^2}{2mr^2} - \frac{GMm}{r}$$

where the last two terms constitute the effective potential.

- $\frac{dV_{eff}}{dr} = 0$ when $r_{\pm} = \frac{J^2}{2M} \left[1 \pm \sqrt{1 - 12 \left(\frac{M}{J} \right)^2} \right]$; $r_+ = r_{min}$ is where stable circular orbit occurs. (to add: V_{eff} vs. r/M plot, LN11-13P8) Thus, the location of circular orbit depends upon $\frac{J}{M}$. If $\frac{M}{J} > \sqrt{12}$ or $\frac{J}{M} < \sqrt{12}$, we don't have equilibrium points. Moreover, $\frac{J}{M} = \sqrt{12}$ (i.e. $r = 6M$) is the location of the inner most stable circular orbit (ISCO).
- For a timelike geodesic $\mathcal{L} = -1$, rearranging variables gives

$$\left(\frac{dr}{d\phi} \right)^2 = \frac{E^2 r^4}{J^2} - r^2 \left(1 - \frac{2M}{r} \right) \left(\frac{r^2}{J^2} + 1 \right)$$

Integrating gives the precession

$$\Delta\phi = -2r + 2 \int_{r_{min}}^{r_{max}} \frac{Jdr}{r^2 \sqrt{E^2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{J^2}{r^2}\right)}}$$

As can be seen from the figure, $\frac{dr}{d\phi} = 0$ at $r_{max} = r_+$ and $r_{min} = r_-$, thus $\frac{E^2}{J^2} = f(r_{\pm}) \left(\frac{1}{J^2} + \frac{1}{r_{\pm}^2} \right)$. After applying Newtonian approximations, one finds

$$\Delta\phi \approx \frac{6\pi GM}{2} \cdot \left(\frac{1}{r_+} + \frac{1}{r_-} \right) \text{ rad/revolution}$$

- For Mercury, $\frac{1}{2} \left(\frac{1}{r_+} + \frac{1}{r_-} \right) = 55.3 \times 10^6 \text{ km}$ and $M_{\odot}G = 1.475 \text{ km}$ gives $\Delta\phi = 0.1038''$ per revolution. Since Mercury makes 415 revolutions per century, $\Delta\phi = 43.03''$ per century.⁴

⁴See Weinberg's *Gravitation and Cosmology*.

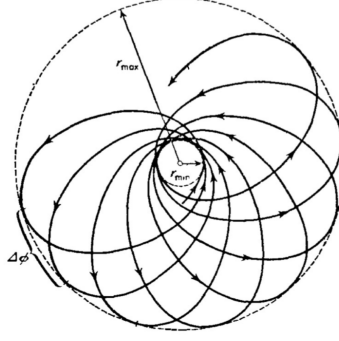


Figure 1: From Landau & Lifshitz Classical Mechanics.

- For a null geodesic $\mathcal{L} = 0$, we can write

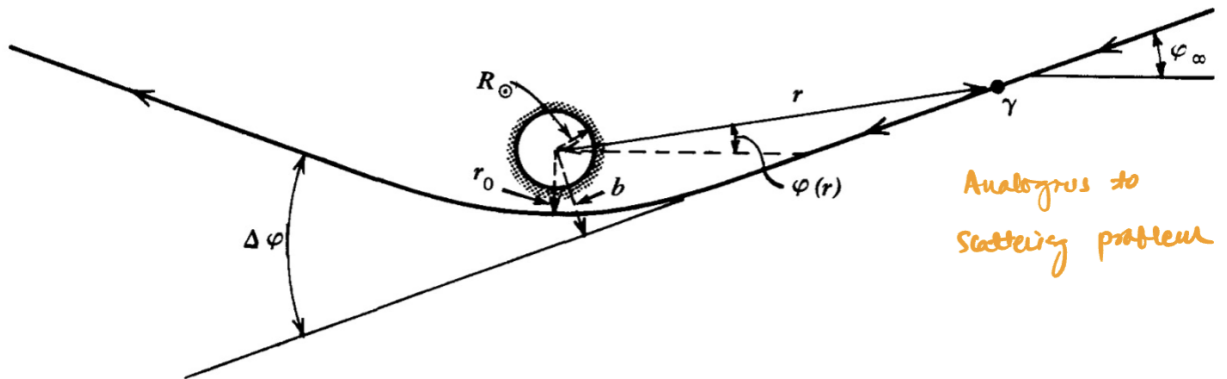
$$\frac{1}{J^2} \left(\frac{dr}{d\lambda} \right)^2 + W_{eff}(r) = \frac{E^2}{J^2} = \frac{1}{b^2}$$

where $\dot{t} = \frac{dt}{d\lambda}$ and $W_{eff} = \frac{1}{r^2} \left(1 - \frac{2M}{r} \right)$. Here the constants of motion $E = f(r)\dot{t}$ and $J = r^2\dot{\phi}$ are energy and angular momentum of the particle following the null geodesic. Basically, $k_\mu = \frac{\partial \mathcal{L}}{\partial x^\mu}$, so $k_t = -E \equiv -\omega = -f(r)\frac{dt}{d\lambda}$ and $k_\phi = J = r^2\frac{d\phi}{d\lambda}$ are conserved. Here, $k_\mu k^\mu = 0$ and k_μ is the wave four-vector (or four-momentum).

- Unlike timelike geodesics where only proper time yields $\mathcal{L} = -1$, any affine parameter λ that is multiplied by a constant gives $\mathcal{L} = 0$ for null geodesics. Thus, only the ratio of E and J determines the shape of the orbit.
- For a null geodesic, rearranging the expression for $\mathcal{L} = 0$ gives

$$\left(\frac{dr}{d\phi} \right)^2 + r^2 - 2Mr - \frac{E^2}{J^2} r^4 = 0$$

At the closest approach $r = r_0$, $\frac{dr}{d\phi} = 0$, so



$$\frac{E^2}{J^2} = \frac{1}{r_0^2} \left(1 - \frac{2M}{r_0} \right)$$

$$\phi = \int \frac{dr/r^2}{\sqrt{f(r_0)/r_0^2 - f(r)/r^2}}$$

By symmetry, $\Delta\phi = \pi - 2|\phi_{r_0} - \phi_\infty|$. For a star, $r_0 \gg 2M = \frac{2GM}{c^2}$, e.g. for the sun $\frac{2GM_\odot}{c^2} \sim 3$ km whereas $R_\odot = 7 \cdot 10^5$ km. Thus, we can take $\frac{2M}{r} \ll 1$ to apply Taylor expansion for approximation. This yields $\delta\phi \approx \frac{4M}{r_0} \approx \frac{4M}{b}$ where b is the impact parameter.

3.9 Gravitational redshift

- Consider an observer moving with unit four-velocity u_o^μ and emitter with u_e^μ . The wave four-vector (or four-momentum) of light follows the null geodesic equation $\frac{dk^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu k^\nu k^\sigma = 0$ with $k^\mu k_\mu = 0$. The frequency at which the emitter transmits light is $\omega_e = -k_\mu u_e^\mu$, and the observer receives light at $\omega_o = -k_\mu u_o^\mu$. The redshift in wavelength then is

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} = \left(\frac{\lambda_o}{\lambda_e} \right) - 1 = \left(\frac{\omega_e}{\omega_o} \right) - 1$$

$$= \left(\frac{k_\mu u_e^\mu}{k_\mu u_o^\mu} \right) - 1$$

- For a stationary observer and emitter in Schwarzschild spacetime, $u_o^k = u_e^k = 0$; from the line element, $-d\tau^2 = g_{tt}dt^2$, so $u_o^0 = \frac{1}{\sqrt{-g_{tt}(o)}}$ and $u_e^0 = \frac{1}{\sqrt{-g_{tt}(e)}}$. Then $-k_\mu u^\mu|_o = \omega_o = \frac{\omega}{\sqrt{-g_{tt}(o)}}$ and $\omega_e = \frac{\omega}{\sqrt{-g_{tt}(e)}}$, and since $\omega = E = \frac{1}{2} \frac{\partial \mathcal{L}}{\partial t} = -\left(1 - \frac{2M}{r}\right) \dot{t}$ is constant, the redshift is

$$z = \left(\frac{1 - \frac{2M}{r_o}}{1 - \frac{2M}{r_e}} \right)^{1/2} - 1$$

In the weak field limit, $2M \ll r_o$ and r_e , so a first-order Taylor approximation gives us $z \approx M \left(\frac{1}{r_e} - \frac{1}{r_o} \right)$.

- In the previous example, if $r_e < r_o$, then $z > 0$, so as photons climb out of gravitational potential, they lose energy and are redshifted. Similarly, in the opposite case $r_o < r_e$, photons gain energy and are blueshifted as they follow the gravitational potential well.

3.10 Horizon and black holes

- The Schwarzschild metric is singular at $r = 2M$ or $r = \frac{2GM}{c^2}$. For the sun, $r = 2M_\odot \sim 3$ km, so the singularity is well within the star where the solution is not vacuum anymore. But when a star is massive enough, then the gravitational collapse when the star runs out of fuel can lead to objects for which this singularity becomes relevant.
- The Schwarzschild coordinates are valid from $2M$ to ∞ . To cover $r < 2M$, one has to find different coordinates that cover all of spacetime.

4 Wormholes

A *wormhole* is a compact region of spacetime whose boundary is topologically trivial, but whose interior is not simply connected.

- For this section, unless otherwise stated, consider the following spherically symmetric and static wormhole solution:

$$ds^2 = -e^{2\Phi(r)} dt^2 + \frac{dr^2}{1 - b(r)/r} + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

$\Phi(r)$ is called the *redshift function*, and $b(r)$ is called the *shape function*, as it determines the shape of the wormhole.

- The radial coordinate r is non-monotonic in that it decreases from $+\infty$ to a minimum value r_0 representing the location of the *throat* of the wormhole, where $b(r_0) = r_0$, and then increases from r_0 to $+\infty$.
- Although the metric coefficient g_{rr} diverges at the throat (as signalled by the coordinate singularity), the proper radial distance $\ell(r) = \pm \int_{r_0}^r [1 - b(r)/r]^{-1/2} dr$ is required to be finite everywhere. The proper distance increases from $\ell = +\infty$ (in the upper universe) to $\ell = 0$ (at the throat) and then from zero to $-\infty$ (in the lower universe).
- In order for the wormhole to be traversable, there must not be any horizons, i.e. that $g_{tt} = -e^{2\Phi(r)} \neq 0$, so $\Phi(r)$ must be finite everywhere. Moreover, whether the wormhole is attractive/repulsive depends on the sign of $\Phi' := \frac{d\Phi}{dr}$, as $a^r = \Phi' (1 - \frac{b}{r})$.
- Much useful information for the choice of the shape function can be extracted from using embedding diagrams. For example, by symmetry we can consider an equatorial slice $\theta = \pi/2$ without loss of generality, and a fixed moment in time; embedding this metric into Euclidean 3-space and writing in cylindrical coordinates (r, ϕ, z) , the embedding surface is given by $\frac{dz}{dr} = \pm \left(\frac{r}{b(r)} - 1 \right)^{-1/2}$. To be a solution of a wormhole, we must impose that the throat *flares out*; i.e. that the inverse of the embedding function, $r(z)$, must satisfy $\frac{d^2 r}{dz^2} = \frac{b - b'r}{2b^2} > 0$ at or near the throat, where $b' := \frac{db}{dr}$. At the throat we verify that the form function satisfies $b'(r_0) < 1$. **This treatment has the drawback of being coordinate dependent (add covariant flaring out condition)**
- Let us work in a reference frame of a set of observers who remain at rest in (t, r, θ, ϕ) , defining the orthonormal basis vectors $e_{\hat{t}} = e^{-\Phi} e_t$, $e_{\hat{r}} = (1 - b/r)^{1/2} e_r$, $e_{\hat{\theta}} = r^{-1} e_\theta$, $e_{\hat{\phi}} = (r \sin \theta)^{-1} e_\phi$. In this basis, the stress-energy tensor is diagonal, with $T_{\hat{t}\hat{t}} = \rho(r)$, $T_{\hat{r}\hat{r}} = -\tau(r)$, $T_{\hat{\theta}\hat{\theta}} = T_{\hat{\phi}\hat{\phi}} = p(r)$ where $\rho(r)$ is the energy density, $\tau(r)$ is the radial tension (with $\tau(r) = -p_r(r)$), and $p(r)$ is the pressure measured in the tangential directions (orthogonal to the radial direction). The Einstein field equations read

$$\begin{aligned} \rho(r) &= \frac{1}{8\pi} \frac{b'}{r^2} \\ \tau(r) &= \frac{1}{8\pi} \left[\frac{b}{r^3} - 2 \left(1 - \frac{b}{r} \right) \frac{\Phi'}{r} \right] \\ p(r) &= \frac{1}{8\pi} \left(1 - \frac{b}{r} \right) \left[\Phi'' + (\Phi')^2 - \frac{b'r - b}{2r^2(1 - b/r)} \Phi' - \frac{b'r - b}{2r^3(1 - b/r)} + \frac{\Phi'}{r} \right] \end{aligned}$$

- Note that the sign of energy density depends on the sign of $b'(r)$. Although wormholes need not necessarily be threaded by negative energy densities or negative matter, by the flaring out condition observers traversing the wormhole with sufficiently high velocities will measure a negative energy density.
- Following from the field equations, relativistic Euler equation, or hydrostatic equation for equilibrium for the material threading the wormhole, reads

$$\tau' = (\rho - \tau)\Phi' - \frac{2}{r}(p + \tau)$$

- Define the dimensionless *exoticity function*

$$\xi = (\tau - \rho)/|\rho| = \frac{b/r - b' - 2r(1 - b/r)\Phi'}{|b'|} = \frac{2b^2}{r|b'|} \frac{d^2r}{dz^2} - 2r \left(1 - \frac{b}{r}\right) \frac{\Phi'}{|b'|}$$

Taking into account the finite character of ρ , and consequently of b' , and the fact that $(1 - b/r)\Phi' \rightarrow 0$ at the throat, we have $\xi(r_0) = \frac{\tau_0 - \rho_0}{|\rho_0|} > 0$, i.e. $\tau_0 > \rho_0$. The latter states that the radial tension at the throat should exceed the energy density. Matter constrained by this condition is called *exotic matter*. Indeed, applying a Lorentz boost in the z -direction, $T_{\hat{0}'\hat{0}'} = \gamma^2(\rho_0 - v^2\tau_0)$; for sufficiently high velocities, $T_{\hat{0}'\hat{0}'} < 0$. This holds for any traversable, nonspherical, and nonstatic wormhole.

- The *null energy condition (NEC)* asserts that for any null vector k^μ , we have $T_{\mu\nu}k^\mu k^\nu \geq 0$. For a diagonal stress-energy tensor, this implies $\rho - \tau \geq 0$ and $\rho + p \geq 0$. Although classical forms of matter are believed to obey the energy conditions, they are violated by certain quantum fields. The Casimir effect is an example. In particular, the flaring-out condition entails the violation of the NEC at the throat.

5 Generalizing spacetime geometry

The notions discussed in the second section transfer quite straightforwardly to the general formalism of differential geometry.⁵ In sections 2 and 3, we study GR in a *Riemannian geometry* $(\mathcal{M}, g)$, where $\mathcal{M}$ is the spacetime manifold and g is the metric. Because we have a notion of length and distance, it is possible to talk about its geodesics, magnitudes of vectors, areas, volumes, and so on. Given a metric, one can compute its Christoffel symbols, AKA its Levi-Civita connection. Thus, a Riemannian manifold comes naturally equipped with a covariant derivative. Recall that we stated that in GR, the covariant derivative satisfies a torsion-free condition. Another property of the Levi-Civita connection of the metric is that it is metric-compatible, a notion that we will discuss in this section. Therefore, Riemannian geometries are characterized by having curvature, but no torsion and no non-metricity. To define a covariant derivative on a manifold, no metric is needed in general: the covariant derivative is instead more generally induced by an affine connection Γ .⁶ The most general type of geometry is *metric-affine geometry*, consisting of the triple (M, g, Γ) . We

⁵I generally skip the details of differential geometry here.

⁶Recall that an affine connection Γ is defined with respect to a basis $\{e_\mu\}$ of $T\mathcal{M}$ by $\nabla_{e_\mu} e_\nu = \Gamma_{\mu\nu}^\alpha e_\alpha$, where ∇ is the covariant derivative satisfying the axioms stated in section 2; then, we can read off the component expression for the covariant derivative of a vector field A , which is $\nabla_\mu A^\nu = \partial_\mu A^\nu + \Gamma_{\mu\lambda}^\nu A^\lambda$. This expression should look familiar from subsection 2.2. Indeed, the expressions in 2.2 and 2.3 transfer perfectly to the case of affine geometry, where the Levi-Civita connection is replaced with an arbitrary affine connection.

subdivide metric-affine geometries by their curvatures, torsions, and non-metricities, which are notions which we define in the subsection below.

We add as a note that the classification of vectors as spacelike/timelike/null can be extended to a curve, which is spacelike/null/timelike if its tangent vector is everywhere spacelike/null/timelike, and to a hypersurface, which is spacelike/null/timelike if its normal vector is everywhere time-like/null/spacelike.

5.1 Curvature, torsion, and non-metricity

- As mentioned in the preceding footnote, the definition of the curvature tensor is the same in metric-affine geometry as in Riemannian geometry.
- Recall that curvature can be visualized as a measurement of the change of direction for a vector that is parallel transported around a closed loop. The notion of *torsion* can also be visualized in terms of parallel transport. Consider $A^\mu, B^\mu \in T_p\mathcal{M}$ defined at a point $p \in \mathcal{M}$, and consider parallel transporting A^μ along B^μ to its tip, and conversely parallel transporting B^μ along A^μ to its tip. In Euclidean geometry, this looks like a parallelogram: the tips of A^μ (resp. B^μ) after parallel transporting along its counterpart coincide. However, when a generic connection is being used to define parallel transport, the same operation described above (at an infinitesimal level) no longer leads to a closed parallelogram. Rather, the two parallel transports end on different points; the difference between the vectors describing their positions relative to p is measured by the torsion tensor. Concretely, defining C'^μ to be the vector from p to the tip of A^μ after parallel transporting along B^μ and defining the other vector C^μ in the obvious similar way, $C'^\alpha - C^\alpha = (\Gamma_{\mu\nu}^\alpha - \Gamma_{\nu\mu}^\alpha)\delta B^\mu\delta A^\nu$, so we define the *torsion tensor* to be the anti-symmetric part of the connection

$$T^\alpha{}_{\mu\nu} = \Gamma_{\mu\nu}^\alpha - \Gamma_{\nu\mu}^\alpha$$

- Consider a curve γ and let B^μ be the tangent vector to γ . If $A \in T\mathcal{M}$, the parallel transport equation $B^\alpha\nabla_\alpha A^\mu = 0$ allows us to determine how the magnitude changes under parallel transport. Recall that the magnitude is defined by the metric as $\|A\|^2 = g(A, A) = g_{\mu\nu}A^\mu A^\nu$. By Leibniz's rule, $B^\alpha\nabla_\alpha(g_{\mu\nu}A^\mu A^\nu) = (B^\alpha\nabla_\alpha g_{\mu\nu})A^\mu A^\nu + 2g_{\mu\nu}(B^\alpha\nabla_\alpha A^\mu)A^\nu$. The last term vanishes due to the parallel transport equation, and we introduce the *non-metricity tensor*

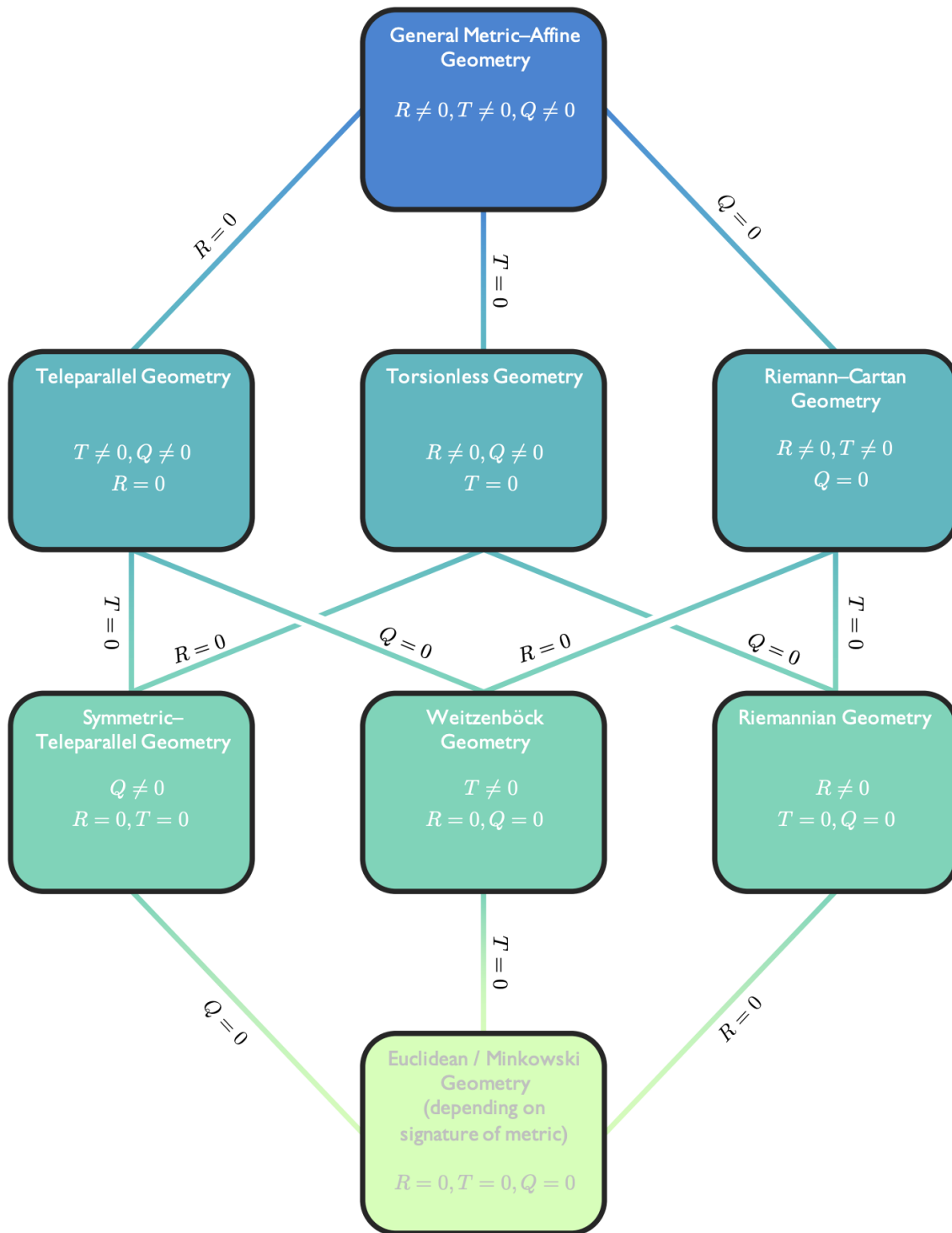
$$Q_{\alpha\mu\nu} = \nabla_\alpha g_{\mu\nu}$$

so that the change $B^\alpha\nabla_\alpha(g_{\mu\nu}A^\mu A^\nu) = Q_{\alpha\mu\nu}B^\alpha A^\mu A^\nu$. The condition $Q_{\alpha\mu\nu} = 0$ is called the *metricity condition* and a connection which satisfies it is called *metric-compatible*.

- Since the non-metricity tensor has three indices, two of which are symmetric, we can define two independent traces $Q_\alpha = Q_{\alpha}{}^\mu{}_\mu$ and $\bar{Q}_\alpha = Q^\mu{}_{\mu\alpha}$.

5.2 Metric-affine geometry

As mentioned in the introduction of this section, we can subdivide metric-affine geometry by whether R, T, Q individually vanish. When all vanish, we have a flat spacetime.



- In a generic metric-affine geometry $(\mathcal{M}, g, \Gamma)$, the connection coefficients can be decomposed as

$$\Gamma_{\mu\nu}^{\alpha} = \left\{ \begin{matrix} \alpha \\ \mu\nu \end{matrix} \right\} + K^{\alpha}_{\mu\nu} + L^{\alpha}_{\mu\nu}$$

$$\text{where } K^{\alpha}_{\mu\nu} = \frac{1}{2}T^{\alpha}_{\mu\nu} + T_{(\mu}{}^{\alpha}{}_{\nu)}$$

$$\text{and } L^{\alpha}_{\mu\nu} = \frac{1}{2}Q^{\alpha}_{\mu\nu} - Q_{(\mu}{}^{\alpha}{}_{\nu)}$$

where we have introduced the *contorsion tensor* $K^{\alpha}_{\mu\nu}$ and the *disformation tensor* $L^{\alpha}_{\mu\nu}$, recalling in their definitions that the brackets in the indices denote the antisymmetrization, and where $\left\{ \begin{matrix} \alpha \\ \mu\nu \end{matrix} \right\} = \frac{1}{2}g^{\alpha\lambda}(\partial_{\mu}g_{\nu\lambda}\partial_{\nu}g_{\mu\lambda} - \partial_{\lambda}g_{\mu\nu})$ are the Christoffel symbols, aka Levi-Civita connection. From this decomposition, one recovers the fact that a torsion-free, metric-compatible connection is uniquely given by the Levi-Civita connection⁷.

- In Riemannian geometry, one can replace the computation of the Lie derivative the coordinate derivative ∂_{μ} by the covariant derivative⁸ $\mathcal{D}_{\mu}$ without altering the result, since all terms containing a connection cancel out at the end of the computation. It is sometimes useful, for instance when discussing symmetries of metric-affine geometries, to compute the Lie derivative in terms of a general affine connection:

$$\mathcal{L}_A g_{\mu\nu} = 2g_{\lambda(\mu}\nabla_{\nu)}A^{\lambda} + (Q_{\lambda\mu\nu} - 2T_{(\mu\nu)\lambda})A^{\lambda} = 2\mathcal{D}_{(\mu}A_{\nu)}$$

$$\mathcal{L}_A \Gamma_{\mu\nu}^{\alpha} = \nabla_{\mu}\nabla_{\nu}A^{\alpha} - T^{\alpha}_{\nu\lambda}\nabla_{\mu}A^{\lambda} - (R^{\alpha}_{\nu\mu\lambda} + \nabla_{\mu}T^{\alpha}_{\nu\lambda})A^{\lambda}$$

- A 1-parameter family of diffeomorphisms $\phi_s : \mathbb{R} \times \mathcal{M} \rightarrow \mathcal{M}$ satisfying $\phi_0 = id$, $\phi_s \circ \phi_t = \phi_{s+t}$, smooth in s , is a *symmetry of a metric-affine geometry* $(\mathcal{M}, g, \Gamma)$ if it satisfies the symmetry condition $\phi_s^* g_{\mu\nu} = g_{\mu\nu}$ and $\phi_s^* \Gamma_{\mu\nu}^{\alpha} = \Gamma_{\mu\nu}^{\alpha}$. We can also consider the infinitesimal symmetry condition, obtained by expanding the original symmetry condition to first order in s around $s = 0$; it reads $\mathcal{L}_{\xi} g_{\mu\nu} = 0$, $\mathcal{L}_{\xi} \Gamma_{\mu\nu}^{\alpha} = 0$, with $\xi := \frac{d}{ds}\phi_s \Big|_{s=0}$ the *generating vector field* which generates the flow ϕ_s .
- The generalized Gauss' theorem for a generic metric-affine geometry $(\mathcal{M}, g, \Gamma)$ takes the form⁹

$$\int_{\mathcal{M}} d^4x \sqrt{|g|} \nabla_{\mu} A^{\mu} = \oint_{\partial\mathcal{M}} d^3y \sqrt{|h|} \varepsilon n_{\mu} A^{\mu} - \int_{\mathcal{M}} d^4x \sqrt{|g|} \left(\frac{1}{2} Q_{\mu} + T_{\mu} \right) A^{\mu}$$

where n^{μ} is the outward-pointing normal vector to $\partial\mathcal{M}$, $\varepsilon = n_{\mu}n^{\mu} = \pm 1$, and h is the determinant of the induced metric on $\partial\mathcal{M}$ obtained by pulling back $g_{\mu\nu}$ to $\partial\mathcal{M}$. Equivalently,

$$\int_{\mathcal{M}} d^4x \nabla_{\mu} (\sqrt{|g|} A^{\mu}) = \oint_{\partial\mathcal{M}} d^3y \sqrt{|h|} \varepsilon n_{\mu} A^{\mu} - \int_{\mathcal{M}} d^4x \sqrt{|g|} T_{\mu} A^{\mu}$$

⁷From this point, we use $\Gamma_{\mu\nu}^{\alpha}$ to denote a general affine connection, reserving $\left\{ \begin{matrix} \alpha \\ \mu\nu \end{matrix} \right\}$ for the Levi-Civita connection.

⁸From this point, we reserve the symbol $\mathcal{D}$ to denote the covariant derivative with respect to the Levi-Civita connection, and ∇ to denote the covariant derivative with respect to a general affine connection

⁹From now on, we use $|g|$ instead of $-g$, since we used $-g$ to ensure that $\sqrt{-g}$ is real when using a $(-, +, +, +)$ signature; on the other hand, using $|g|$ instead makes no assumption about the signature.

- Adding a $(1, 2)$ tensor to a given connection $\Gamma_{\mu\nu}^\alpha$ results in a new and equally valid connection $\hat{\Gamma}_{\mu\nu}^\alpha = \Gamma_{\mu\nu}^\alpha + \Omega_{\mu\nu}^\alpha$. We can then compute the curvature tensor of the connection $\hat{\Gamma}_{\mu\nu}^\alpha$ and express it in terms of the curvature of $\Gamma_{\mu\nu}^\alpha$ as well as contributions from $\Omega_{\mu\nu}^\alpha$:

$$\hat{R}^\alpha_{\beta\mu\nu} = R^\alpha_{\beta\mu\nu} + T^\lambda_{\mu\nu}\Omega^\alpha_{\lambda\beta} + 2\mathcal{D}_{[\mu}\Omega^\alpha_{\nu]\beta} + 2\Omega^\alpha_{[\mu|\lambda|}\Omega^\lambda_{\nu]\beta}$$

Thus, we can find a relationship between the curvature tensor $R^\alpha_{\mu\nu\rho}(\Gamma)$ of a general affine connection and the curvature tensor $\mathcal{R}^\alpha_{\mu\nu\rho}(g)$ of the Levi-Civita connection; simply replacing the original connection with the Levi-Civita connection, and $\Omega^\alpha_{\mu\nu}$ with $K^\alpha_{\mu\nu} + L^\alpha_{\mu\nu}$,

$$\begin{aligned} R^\alpha_{\mu\nu\rho}(\Gamma) &= R^\alpha_{\mu\nu\rho}(g) + T^\lambda_{\nu\rho}K^\alpha_{\lambda\mu} + 2D_{[\nu}K^\alpha_{\rho]\mu} + T^\lambda_{\nu\rho}L^\alpha_{\lambda\mu} + 2D_{[\nu}L^\alpha_{\rho]\mu} \\ &\quad + 2K^\alpha_{[\nu|\lambda}K^\lambda_{\rho]\mu} + 2L^\alpha_{[\nu|\lambda}K^\lambda_{\rho]\mu} + 2K^\alpha_{[\nu|\lambda}L^\lambda_{\rho]\mu} + 2L^\alpha_{[\nu|\lambda}L^\lambda_{\rho]\mu} \end{aligned}$$

For the Ricci scalars of the two connections,

$$R(\Gamma) = \mathcal{R}(g) + \mathbb{T} + \mathbb{Q} + T^{\rho\mu\nu}Q_{\mu\nu\rho} - T^\mu Q_\mu + T^\mu \bar{Q}_\mu + \mathcal{D}_\alpha(Q^\alpha - \bar{Q}^\alpha + 2T^\alpha)$$

where we have introduced the *torsion scalar* and *non-metricity scalar*, respectively given by

$$\begin{aligned} \mathbb{T} &= \frac{1}{2} \left(\frac{1}{4} T_{\alpha\mu\nu} + \frac{1}{2} T_{\mu\alpha\nu} - g_{\alpha\mu} T_\nu \right) T^{\alpha\mu\nu} \\ \mathbb{Q} &= \frac{1}{4} Q_{\alpha\mu\nu} Q^{\alpha\mu\nu} - \frac{1}{2} Q_{\alpha\mu\nu} Q^{\mu\alpha\nu} - \frac{1}{4} Q_\alpha Q^\alpha + \frac{1}{2} Q_\alpha \bar{Q}^\alpha \\ &= g^{\mu\nu} \left(L^\alpha_{\alpha\beta} L^\beta_{\mu\nu} - L^\alpha_{\beta\mu} L^\beta_{\nu\alpha} \right) \end{aligned}$$

For zero non-metricity,

$$R(\Gamma) = \mathcal{R}(g) + \mathbb{T} + 2\mathcal{D}_\alpha T^\alpha$$

and for vanishing torsion,

$$R(\Gamma) = \mathcal{R}(g) + \mathbb{Q} + \mathcal{D}_\alpha(Q^\alpha - \bar{Q}^\alpha)$$

- We now list identities for metric-affine geometries.

$$\begin{aligned} R^\alpha_{\mu(\nu\rho)} &= 0 \\ R^\mu_{[\alpha\beta\gamma]} - \nabla_{[\alpha} T^\mu_{\beta\gamma]} + T^\lambda_{[\alpha\beta} T^\mu_{\gamma]\lambda} &= 0 \\ \nabla_{[\alpha} R^\mu_{\nu|\beta\gamma]} - T^\lambda_{[\alpha\beta} R^\mu_{\nu|\gamma]\lambda} &= 0 \end{aligned}$$

Let f be a scalar field and ω_α a 1-form.

$$\begin{aligned} [\nabla_\mu, \nabla_\nu]f &= -T^\lambda_{\mu\nu}\partial_\lambda f \\ [\nabla_\mu, \nabla_\nu]A^\alpha &= R^\alpha_{\lambda\mu\nu}A^\lambda - T^\lambda_{\mu\nu}\nabla_\lambda A^\alpha \\ [\nabla_\mu, \nabla_\nu]\omega_\alpha &= -R^\lambda_{\alpha\mu\nu}\omega_\lambda - T^\lambda_{\mu\nu}\nabla_\lambda\omega_\alpha \\ [\nabla_\mu, \nabla_\nu]g_{\alpha\beta} &= 2\nabla_{[\mu}Q_{\nu]\alpha\beta} = -2R_{(\alpha\beta)\mu\nu} - T^\lambda_{\mu\nu}Q_{\lambda\alpha\beta} \\ [\nabla_\mu, \nabla_\nu]|g|^{w/2} &= \frac{w}{2}|g|^{w/2}g^{\alpha\beta}[\nabla_\mu, \nabla_\nu]g_{\alpha\beta} = w|g|^{w/2}\nabla_{[\mu}Q_{\nu]\alpha\beta} = w|g|^{w/2}\nabla_{[\mu}Q_{\nu]} \end{aligned}$$

For tensors and tensor densities, we have

$$\begin{aligned} [\nabla_\mu, \nabla_\nu]A^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_q} &= R^{\mu_1}_{\lambda\mu\nu}A^{\lambda\cdots\mu_p}_{\nu_1\cdots\nu_q} + \cdots + R^{\mu_p}_{\lambda\mu\nu}A^{\mu_1\cdots\lambda}_{\nu_1\cdots\nu_q} \\ &\quad - R^\lambda_{\nu_1\mu\nu}A^{\mu_1\cdots\mu_p}_{\lambda\cdots\nu_q} - \cdots - R^\lambda_{\nu_q\mu\nu}A^{\mu_1\cdots\mu_p}_{\nu_1\cdots\lambda} \\ &\quad - T^\lambda_{\mu\nu}\nabla_\lambda A^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_q} \\ [\nabla_\mu, \nabla_\nu] \left(|g|^{w/2} A^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_q} \right) &= |g|^{w/2} ([\nabla_\mu, \nabla_\nu]A^{\mu_1\cdots\mu_p}_{\nu_1\cdots\nu_q} + w\nabla_{[\mu}Q_{\nu]}) \end{aligned}$$

5.3 Einstein's general relativity revisited and the Palatini formalism

We start with a metric-affine geometry $(\mathcal{M}, g, \Gamma)$. Neither $\mathcal{M}$ nor g are fixed; both will be determined as solutions of Einstein's field equations. We postulate that Γ is torsion-free and metric-compatible, which leave the curvature tensor $\mathcal{R}_{\mu\nu\rho}^\alpha$ as the only non-zero tensor characterizing the spacetime geometry. Again, the geometric postulates of torsion-freeness and metric-compatibility are satisfied if and only if the connection is the Levi-Civita connection. Because the Levi-Civita connection is completely determined by the metric, we will omit Γ from the triple and simply say that spacetime is modelled by the pair $(\mathcal{M}, g)$.

- The action functional defining GR is the Einstein-Hilbert (EH) action plus the Gibbons-Hawking-York (GHY) boundary term, which we omitted in our first treatment because we were solely interested in deriving the bulk Einstein field equations. This term, however, is necessary whenever $\mathcal{M}$ has a boundary $\partial\mathcal{M}$, otherwise the variational principle is ill-defined. Including a cosmological term and a matter action for completeness,

$$\begin{aligned}\mathcal{S}_{\text{GR}}[g, \Psi] &= \mathcal{S}_{\text{EH}}[g] + \mathcal{S}_{\text{matter}}[g, \Psi] \\ &= \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} (\mathcal{R} - 2\Lambda) + \frac{1}{\kappa} \oint_{\partial\mathcal{M}} d^3y \sqrt{|h|} \varepsilon \mathcal{K} + \mathcal{S}_{\text{matter}}[g, \Psi]\end{aligned}$$

with $\kappa = 8\pi G$. $\mathcal{S}_{\text{matter}}[g, \Psi]$ is the action of tensorial matter fields Ψ which are minimally coupled to the gravitational field $g_{\mu\nu}$. Spinorial fields are not described by this action; to describe fermions, we would have to describe gravity in terms of a tetrad field rather than a metric tensor. Note also that $\mathcal{K}$ is the trace of the extrinsic curvature of $\partial\mathcal{M}$ viewed as a hypersurface embedded into $\mathcal{M}$.

- For our purposes moving forwards, we define the energy-momentum tensor of matter fields in full generality

$$\mathcal{T}_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta \mathcal{S}_{\text{matter}}}{\delta g^{\mu\nu}}$$

- Given a set of initial data on a three-dimensional Cauchy surface, the Einstein field equations $\mathcal{R}_{\mu\nu} - \frac{1}{2}\mathcal{R}g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$ determine the metric and the manifold up to diffeomorphisms, analogous to how Maxwell's equations determine the vector potential up to gauge transformations.

We can actually formulate Einstein's GR in the framework of general metric-affine geometry $(\mathcal{M}, g, \Gamma)$, where Γ a priori has non-trivial curvature, torsion, and non-metricity. In the *Palatini formalism*, metric and connection are treated as two independent fields and the action is varied with respect to both. Even if we start with a completely general Γ , the field equations fix Γ to be the Levi-Civita connection up to a projective symmetry!

- The Palatini action functional for GR in absence of a cosmological constant is

$$\mathcal{S}_{\text{GR}}[g, \Gamma] = \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} g^{\mu\nu} R_{\mu\nu}(\Gamma) + \mathcal{S}_{\text{matter}}$$

We do not need to include a boundary term, since there are no second order derivatives.

- Varying this action with respect to the inverse metric gives the metric field equations

$$R_{(\mu\nu)}(\Gamma) - \frac{1}{2}R(\Gamma)g_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$$

- Now, we vary the Palatini action with respect to Γ while keeping the metric fixed; the variation of the Ricci tensor is given by the Palatini identity $\delta_\Gamma R_{\mu\nu}(\Gamma) = \nabla_\alpha \delta \Gamma_{\nu\mu}^\alpha - \nabla_\nu \delta \Gamma_{\alpha\mu}^\alpha - T^\alpha_{\beta\nu} \delta \Gamma_{\alpha\mu}^\beta$. After performing the calculations and enforcing the standard boundary condition $\delta \Gamma|_{\partial\mathcal{M}} = 0$, we get the *connection field equations*

$$\nabla_\alpha \left(\sqrt{|g|} g^{\mu\nu} \right) - \delta_\alpha^\mu \nabla_\beta \left(\sqrt{|g|} g^{\beta\nu} \right) = \sqrt{|g|} \left[g^{\mu\nu} T_\alpha + g^{\beta\nu} T^\mu_{\alpha\beta} - \delta_\alpha^\mu g^{\beta\nu} T_\beta \right] + \tilde{\mathcal{H}}^{\mu\nu}_\alpha$$

where we have introduced the *hypermomentum of matter*

$$\tilde{\mathcal{H}}^{\mu\nu}_\alpha = 2\kappa \frac{\delta \mathcal{S}_{\text{matter}}}{\delta \Gamma_{\mu\nu}^\alpha}$$

Fermionic fields naturally couple to torsion and therefore contribute to the hypermomentum. If torsion is present, bosonic fields also do contribute to the hypermomentum.

- It can be found that the connection field equations are purely algebraic equations for the connection which can be completely solved, and fix the connection to the Levi-Civita connection up to a projective transformation $\Gamma_{\mu\nu}^\alpha \mapsto \Gamma_{\mu\nu}^\alpha + \delta_\nu^\alpha \xi_\mu$.

The lesson to be taken from the above is that as long as the action has the same form as the EH action, changing the geometric framework will not lead to a new formulation of GR. We return to studying the EH action.

- The integrand of the EH action, $\sqrt{|g|}\mathcal{R}$, is a scalar density of weight 1. Therefore, the EH action is invariant under diffeomorphisms. This invariance gives rise to the *Bianchi identities* $\mathcal{D}_\mu G^{\mu\nu} = 0$, where $G^{\mu\nu} = \mathcal{R}_{\mu\nu} - \frac{1}{2}\mathcal{R}g_{\mu\nu}$ is the Einstein tensor.
- Recall that the metric has ten independent components. Since Einstein's equations are generally covariant, we can perform diffeomorphisms; each diffeomorphism provides us with four choices, which in turn grants us the freedom to fix four components of the metric. The Bianchi identities tell us that four of Einstein's field equations are actually constraints, rather than dynamical equations. Therefore, GR has 2 physical propagating DOFs per spacetime point, i.e. two gravitational wave polarizations.

5.4 The teleparallel equivalent of general relativity¹⁰

TEGR attributes the effects of gravity to a non-vanishing torsion tensor. Starting from a metric-affine geometry $(\mathcal{M}, g, \Gamma)$, the geometric postulates ofTEGR demand that the connection be flat and metric-compatible; unlike the postulates in GR, this does not fix a connection.

- Observe that the trivial connection $\hat{\Gamma}_{\mu\nu}^\alpha = 0$ is obviously flat. under a change of coordinates, a connection transforms inhomogeneously; for a trivial connection, the change of coordinates $\hat{x}^\mu \mapsto x^\mu(\hat{x})$ leads to $\hat{\Gamma}_{\mu\nu}^\alpha \mapsto \Gamma_{\mu\nu}^\alpha = \frac{\partial x^\alpha}{\partial \hat{x}^\lambda} \partial_\mu \frac{\partial \hat{x}^\lambda}{\partial x^\nu}$. Reading off $\frac{\partial \hat{x}^\mu}{\partial x^\nu}$ as the components of a matrix Λ , we rewrite this as $\Gamma_{\mu\nu}^\alpha = (\Lambda^{-1})^\alpha_\lambda \partial_\mu \Lambda_\nu^\lambda$. If the curvature tensor is zero in one coordinate system, it is zero in any other coordinate system; since it is zero for the trivial connection, it is also zero for the connection $\Gamma_{\mu\nu}^\alpha$ above, since this connection has been obtained by a change of coordinates. We conclude that any connection of the form $\Gamma_{\mu\nu}^\alpha = (\Lambda^{-1})^\alpha_\lambda \partial_\mu \Lambda_\nu^\lambda$ where $\Lambda_\nu^\mu \in GL(4, \mathbb{R})$ is flat.

¹⁰The curvature tensor measures the change in direction of a vector which is being parallel transported around a closed loop. Hence, when curvature vanishes, there is no change in direction and the vector remains, in this sense, parallel to itself, justifying the use of the term *teleparallel*.

- Plugging in this flat connection into $Q_\alpha^{\mu\nu} = 0$, we find that $g^{\lambda(\mu}\partial_\alpha\Lambda_\rho^{\nu)}(\Lambda^{-1})_\lambda^\rho = \frac{1}{2}\partial_\alpha g^{\mu\nu}$, allowing us to eliminate the metric and express it in terms of the Γ_ν^μ . This turns the metric into an auxiliary field. The metric has ten components while Γ_ν^μ has 16 components. Six of the components of Γ_ν^μ reflect the freedom to perform local Lorentz transformations.
- The torsion tensor is given by $T^\alpha_{\mu\nu} = 2(\Lambda^{-1})^\alpha_\lambda\partial_{[\mu}\Lambda_{\nu]}^\lambda$ for any flat connection parametrized by $\Lambda \in GL(4, \mathbb{R})$.
- To construct an action, recall that the Ricci scalar of a metric-compatible connection is related to the Ricci scalar of the Levi-Civita connection via $R(\Gamma) = \mathcal{R}(g) + \mathbb{T} + 2\mathcal{D}_\alpha T^\alpha$; imposing flatness, $\mathcal{R}(g) = -\mathbb{T}(\Lambda) - 2\mathcal{D}_\alpha T^\alpha(\Lambda)$, where the metric has been integrated out from the metricity condition. Replacing the Ricci scalar in the EH action with the RH side of this equation yields a strictly equal action. However, dropping $\mathcal{D}_\alpha T^\alpha$, which amounts to a boundary term, yields an action which is genuinely different but leads to the same field equations. The action of TEGR is thus defined

$$\mathcal{S}_{\text{TEGR}}[\Lambda] = -\frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} \mathbb{T}(\Lambda) + \mathcal{S}_{\text{matter}}$$

where it is understood that the metric can be expressed in terms of Λ . Of course, since Λ has 16 components, we find more field equations than in GR; ten of these equations are precisely the Einstein equations, while the remaining six are Bianchi identities related to the local Lorentz symmetry expressed through six of the components of Λ . Note also that dropping $\mathcal{D}_\alpha T^\alpha$ makes the action function depend only on first-order derivatives, removing the need for a boundary term in the action for well-definedness of the variational principle.

- We can also write this action in a covariant form:

$$\mathcal{S}_{\text{TEGR}}[g, \Lambda; \tilde{\Pi}, \tilde{\chi}] = - \int_{\mathcal{M}} d^4x \left(\frac{1}{2\kappa} \sqrt{|g|} \mathbb{T} + \tilde{\Pi}_\alpha^{\mu\nu\rho} R_{\mu\nu\rho}^\alpha + \tilde{\chi}^\alpha_{\mu\nu} Q_\alpha^{\mu\nu} \right) + \mathcal{S}_{\text{matter}}$$

where $\tilde{\Pi}_\alpha^{\mu\nu\rho}, \tilde{\chi}^\alpha_{\mu\nu}$ are tensor densities of weight $w = 1$ which act as Lagrange multipliers, enforcing the postulates of vanishing curvature and non-metricities. These possess symmetries $\tilde{\Pi}_\alpha^{\mu\nu\rho} = \tilde{\Pi}_\alpha^{\mu[\nu\rho]}$, $\tilde{\chi}^\alpha_{\mu\nu} = \tilde{\chi}^\alpha_{(\mu\nu)}$ which they inherit from the curvature tensor and non-metricity tensor, respectively.

- Based on the above Palatini action, we can perform four independent variations, which give us the metric field equations

$$(\nabla_\alpha + T_\alpha) S_{(\mu\nu)}^\alpha + t_{\mu\nu} - \frac{1}{2} \mathbb{T} g_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$$

and the connection field equations

$$(\nabla_\alpha + T_\alpha) \left[\sqrt{|g|} S_{[\mu}^\alpha{}_{\nu]} \right] = 0$$

where we have introduced the *torsion conjugate* $S_\alpha^{\mu\nu}$ and the symmetric tensor $t_{\mu\nu}$,

$$\begin{aligned} S_\alpha^{\mu\nu} &= \frac{\partial \mathbb{T}}{\partial T_{\mu\nu}^\alpha} = \frac{1}{4} T_\alpha^{\mu\nu} + \frac{1}{2} T^{[\mu}{}_\alpha{}^{\nu]} - \delta_\alpha^{[\mu} T^{\nu]} \\ t_{\mu\nu} &= \frac{\partial \mathbb{T}}{\partial g^{\mu\nu}} = \frac{1}{2} S_{(\mu}{}^{\lambda\kappa} T_{\nu)\lambda\kappa} - T^{\lambda\kappa}{}_{(\mu} S_{\lambda\kappa|\nu)} \end{aligned}$$

It is also assumed that the hypermomentum density, which enters as $(\nabla_\alpha + T_\alpha)\tilde{\mathcal{H}}_{[\mu}{}^\alpha{}_{\nu]}$ in the field equations is either identically zero (i.e. the matter content and the matter couplings have been chosen such that there is no contribution to this tensor density) or it is covered in the sense that $(\nabla_\alpha + T_\alpha)\tilde{\mathcal{H}}_{\mu}{}^\alpha{}_{\nu} = 0$; this conservation law holds by virtue of the gauge invariance of the matter sector and is confirmed in all standard cases where a non-trivial hypermomentum arises from coupling matter fields to the connection.

- The metric field equations are simply the Einstein field equations in disguise, while the connection field equations arise as Bianchi identities, and carry no dynamical information; they do not determine the metric nor the connection, and they are trivially satisfied.
- Starting from an action of the form $\mathcal{S}[g, \Gamma]$, where Γ is a generic affine connection, and assuming that the action is diffeomorphism invariant, the Bianchi identities in a metric-affine geometry take the form

$$2\mathcal{D}_\mu \tilde{\mathcal{M}}^\mu{}_\lambda - \hat{\nabla}_\nu \hat{\nabla}_\mu \tilde{\mathcal{C}}_\lambda{}^{\mu\nu} + T_{\lambda\nu}^\alpha \hat{\nabla}_\mu \tilde{\mathcal{C}}_\alpha{}^{\mu\nu} + (R^\alpha{}_{\nu\mu\lambda} - T_\mu T^\alpha{}_{\nu\lambda}) \tilde{\mathcal{C}}_\alpha{}^{\mu\nu} = 0$$

where $\tilde{\mathcal{M}}^{\mu\nu} = \frac{\delta \mathcal{S}[g, \Gamma]}{\delta g_{\mu\nu}}$, $\tilde{\mathcal{C}}_\alpha{}^{\mu\nu} = \frac{\delta \mathcal{S}[g, \Gamma]}{\delta \Gamma_{\mu\nu}^\alpha}$, and $\hat{\nabla}_\mu = \nabla_\mu + T_\mu$. Fixing the connection to be flat and metric-compatible, and using the metric field equations $\tilde{\mathcal{M}}_{\mu\nu} = \sqrt{|g|}G_{\mu\nu}$, the Bianchi identities of TEGR are

$$\hat{\nabla}_\nu \hat{\nabla}_\mu \tilde{\mathcal{C}}_\lambda{}^{\mu\nu} - T^\alpha{}_{\lambda\nu} \hat{\nabla}_\mu \tilde{\mathcal{C}}_\alpha{}^{\mu\nu} + T_\mu T^\alpha{}_{\nu\lambda} \tilde{\mathcal{C}}_\alpha{}^{\mu\nu} = 0$$

- We can count degrees of freedom again. For a metric $g_{\mu\nu}$ and a general affine connection $\Gamma_{\mu\nu}^\alpha$ we have 10 + 64 fields; the flatness condition reduces the connection to 16 DOFs, since it can be parametrized by some $\Lambda_\nu^\mu \in GL(4, \mathbb{R})$, and the postulate of vanishing non-metricity can be solved by $g_{\mu\nu} = \Lambda_\mu^\alpha \Lambda_\nu^\beta c_{\alpha\beta}$ where $c_{\alpha\beta}$ is a symmetric, constant tensor, which we naturally choose to be the Minkowski metric $\eta_{\mu\nu}$. This means that we have 16 total DOFs since the metric is also parametrized by Λ_ν^μ . Flat connections possess a gauge symmetry; transformations $\Lambda \mapsto \Lambda \mathcal{U}$, where $\mathcal{U} \in GL(4, \mathbb{R})$, leave the curvature tensor invariant. If the transformation must also respect the postulate of vanishing non-metricity, it must leave the metric invariant, which means $\mathcal{U}_\kappa^\alpha \mathcal{U}_\lambda^\beta \Lambda_\mu^\kappa \Lambda_\nu^\lambda \eta_{\alpha\beta} = \Lambda_\mu^\alpha \Lambda_\nu^\beta \eta_{\alpha\beta} \implies \mathcal{U} \in SO^+(1, 3)$, i.e. that six components of Λ_ν^μ simply represent Lorentz transformations and that these transformations are pure gauge, leaving us with 10 DOFs. Since TEGR is a generally covariant theory and diffeomorphisms remove 2×4 DOFs, we are finally left with only two DOFs.

5.5 The symmetric teleparallel equivalent of general relativity¹¹

We start again with a metric-affine geometry $(\mathcal{M}, g, \Gamma)$, this time restricted by the geometric postulates $R_{\mu\nu\rho}^\alpha = 0, T_{\mu\nu}^\alpha = 0$.

- We again parametrize a flat connection by $\Gamma_{\mu\nu}^\alpha = (\Lambda^{-1})_\lambda^\alpha \partial_\mu \Lambda_\nu^\lambda$, where $\Lambda_\nu^\mu \in GL(4, \mathbb{R})$. The postulate of vanishing torsion can then be rephrased as $\partial_{[\mu} \Lambda_{\nu]}^\alpha = 0$. Thus we can write $\Lambda_\nu^\mu = \partial_\nu \xi^\mu$, where ξ^μ denotes a collection of four arbitrary functions of the coordinates x^μ (not a vector field). Therefore, a flat, torsionless connection can be written as $\Gamma_{\mu\nu}^\alpha = \frac{\partial x^\alpha}{\partial \xi^\lambda} \partial_\mu \partial_\nu \xi^\lambda$ where $\frac{\partial x^\alpha}{\partial \xi^\lambda}$ is the inverse of the Jacobian matrix $\frac{\partial \xi^\lambda}{\partial x^\alpha}$. That is, in any given coordinate system, we can choose four independent functions such that the Jacobian matrix is invertible and we can then construct a flat and torsionless connection.

¹¹The vanishing of torsion implies that the connection $\Gamma_{\mu\nu}^\alpha$ is symmetric in its lower indices, hence the use of the word symmetric.

- Flat and torsionless connections can be set to zero globally by an appropriate choice of connections. If we choose coordinates such that $x^\mu = \xi^\mu$, the connection is exactly equal to zero because $\partial_\mu \partial_\nu \xi^\lambda = 0$; this is known as the *coincident gauge*.
- In STEGR, $\mathcal{R}(g) = -\mathbb{Q} - \mathcal{D}_\alpha(Q^\alpha - \bar{Q}^\alpha)$. Since the second term amounts to a mere boundary term by the generalized Gauss theorem, we can just replace $\mathcal{R}(g)$ in the EH action by $-\mathbb{Q}$ to get the action functional for STEGR:

$$\mathcal{S}_{\text{STEGR}}[g, \xi] = -\frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} \mathbb{Q}(g, \xi) + \mathcal{S}_{\text{matter}}$$

- Employing the Palatini formalism to express the action principle in a manifestly covariant way, the action is defined as

$$\mathcal{S}_{\text{STEGR}}[g, \Gamma] = - \int_{\mathcal{M}} d^4x \left(-\frac{1}{2\kappa} \sqrt{|g|} \mathbb{Q}(g, \Gamma) + \tilde{\Pi}_\alpha^{\mu\nu\rho} R^\alpha_{\mu\nu\rho} + \tilde{\chi}_\alpha^{\mu\nu} T^\alpha_{\mu\nu} \right) + \mathcal{S}_{\text{matter}}$$

with Lagrange multipliers $\tilde{\Pi}_\alpha^{\mu\nu\rho}$, $\tilde{\chi}_\alpha^{\mu\nu}$ tensor densities of weight 1, which inherit the symmetries $\tilde{\Pi}_\alpha^{\mu\nu\rho} = \tilde{\Pi}_\alpha^{\mu[\nu\rho]}$ and $\tilde{\chi}_\alpha^{\mu\nu} = \tilde{\chi}_\alpha^{[\mu\nu]}$ from the curvature and torsion tensors respectively.

- After performing variations, one obtains the metric field equations

$$\frac{2}{\sqrt{|g|}} \nabla_\alpha \left[\sqrt{|g|} P^\alpha_{\mu\nu} \right] + q_{\mu\nu} - \frac{1}{2} \mathbb{Q} g_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$$

where we have introduced the *non-metricity conjugate* $P^\alpha_{\mu\nu}$ and the symmetric tensor $q_{\mu\nu}$, defined by

$$\begin{aligned} P^\alpha_{\mu\nu} &= \frac{1}{2} \frac{\partial \mathbb{Q}}{\partial Q_{\alpha\mu\nu}} = -\frac{1}{4} Q^\alpha_{\mu\nu} + \frac{1}{2} Q_{(\mu}{}^\alpha{}_{\nu)} + \frac{1}{4} g_{\mu\nu} Q^\alpha - \frac{1}{4} (g_{\mu\nu} \bar{Q}^\alpha + \delta_{(\mu}^\alpha Q_{\nu)}) \\ q_{\mu\nu} &= \frac{\partial \mathbb{Q}}{\partial g^{\mu\nu}} P_{(\mu|\alpha\beta} Q_{\nu)}^{\alpha\beta} - 2 P^{\alpha\beta}_{(\nu} Q_{\alpha\beta|\mu)} \end{aligned}$$

Note that $\mathbb{Q} = P_{\alpha\mu\nu} Q^{\alpha\mu\nu}$.

- The variations with respect to the Lagrange multipliers and the general affine connection boil down to a connection field equation of the form

$$\nabla_\mu \nabla_\nu \left(\sqrt{|g|} P^{\mu\nu}{}_\alpha \right) = 0$$

- Just as in TEGR, we have assumed that hypermomentum density vanishes; alternatively, we could have demanded that it is conserved, $\nabla_\mu H_\alpha^{\mu\nu} = 0$.
- The action is manifestly invariant under diffeomorphisms; we find that for STEGR $\mathcal{D}_\mu \mathcal{M}^\mu{}_\nu + \mathcal{C}_\mu = 0$ where $\mathcal{M}_{\mu\nu} = \frac{2}{\sqrt{|g|}} \nabla_\alpha \left[\sqrt{|g|} P^\alpha_{\mu\nu} \right] + q_{\mu\nu} - \frac{1}{2} \mathbb{Q} g_{\mu\nu}$ and $\mathcal{C}_\alpha = \nabla_\mu \nabla_\nu \left(\sqrt{|g|} P^{\mu\nu}{}_\alpha \right)$. $\mathcal{M}_{\mu\nu}$ is simply the expression that appears on the left side of the metric field equations, while $\mathcal{C}_\alpha$ represents the left side of the connection field equations. As a consequence of the Bianchi identities, it follows that if the metric field equations are satisfied, i.e. if $\mathcal{M}_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$, then $\kappa \mathcal{D}_\mu \mathcal{T}_\nu^\mu + \mathcal{C}_\nu = 0$; by invoking covariant conservation of energy-momentum of matter fields, we find that if the metric field equations are satisfied, then $\mathcal{C}_\nu = 0$. In other words, if the metric

field equations are satisfied, then the connection field equations become identities, trivially satisfied and carrying no dynamical information. Since one can show that $\mathcal{M}_{\mu\nu} = G_{\mu\nu}$, one can reach an even stronger conclusion: the Einstein tensor satisfies the Bianchi identity $\nabla_\mu G^\mu_\nu$ also off-shell, i.e. when the Einstein equations are not satisfied. One reaches the conclusion that $\mathcal{C}_\nu = 0$ is always true!

- Since $\mathcal{M}_{\mu\nu}$ is independent of ξ^α (it only knows about the Levi-Civita part of the connection and nothing else), it follows that $\mathcal{M}_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$ are purely equations for the metric. Furthermore, since $\mathcal{C}_\nu = 0$ always, there are no equations which determine the four functions ξ^α , i.e. they remain completely arbitrary. It follows STEGR is equivalent to GR in the sense that both theories have the same field equations and thus the same solution space; they are nevertheless rooted in different mathematical frameworks and used different fields in their formulation, opening the door to conceptual and philosophical differences between the two theories. Moreover, there is a sense in which STEGR is invariant under two copies of the diffeomorphism group. First, its action is manifestly diffeomorphism invariant and its field equations are manifestly general covariant. Secondly, we have the freedom to choose the four functions ξ^α at will; the metric field equations do not depend upon this choice, and there are no dynamical equations which determine the ξ 's. In $f(\mathbb{Q})$ gravity the ξ 's are no longer arbitrary and come with their own dynamical field equations.
- Indeed, STEGR propagates two physical degrees of freedom; by either postulating the vanishing of curvature and torsion or by solving the constraints that arise from the Palatini formulation of the theory, one finds that the connection carries four potential DOFs (the ξ 's); since these remain completely arbitrary and do not constitute anything physical, this leaves us with at most 10 dynamical variables, namely the metric components. Since the metric field equations are simply the Einstein equations, the same counting as in GR assures us that only two of these components represent physical DOFs.

5.6 Coincident general relativity

CGR is STEGR in the coincident gauge.

- Since $\Gamma^\alpha_{\mu\nu} = \left\{ \begin{smallmatrix} \alpha \\ \mu\nu \end{smallmatrix} \right\} + L^\alpha_{\mu\nu} = 0$, $L^\alpha_{\mu\nu} = -\left\{ \begin{smallmatrix} \alpha \\ \mu\nu \end{smallmatrix} \right\}$ (in the coincident gauge). Hence the CGR action takes the form

$$\begin{aligned} \mathcal{S}_{\text{CGR}}[g] &\equiv \mathcal{S}_{\text{STGR}}[g, \Gamma = 0] = \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} g^{\mu\nu} \left(L^\alpha_{\alpha\beta} L^\beta_{\mu\nu} - L^\alpha_{\beta\mu} L^\beta_{\nu\alpha} \right) \\ &= \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} g^{\mu\nu} \left(\left\{ \begin{smallmatrix} \alpha \\ \alpha\beta \end{smallmatrix} \right\} \left\{ \begin{smallmatrix} \beta \\ \mu\nu \end{smallmatrix} \right\} - \left\{ \begin{smallmatrix} \alpha \\ \beta\mu \end{smallmatrix} \right\} \left\{ \begin{smallmatrix} \beta \\ \nu\alpha \end{smallmatrix} \right\} \right) \end{aligned}$$

The first integral is valid in complete generality, while the second one holds only in the coincident gauge. The action is a functional of the metric alone; the connection has been dealt with and even globally trivialized, such that one only has to work with the metric.

- This action is exactly equal to the Einstein action, which is simply the EH action without the second order derivatives; we can see this since $\mathcal{R}_{\mu\nu} = \partial_\alpha \left\{ \begin{smallmatrix} \alpha \\ \nu\mu \end{smallmatrix} \right\} - \partial_\nu \left\{ \begin{smallmatrix} \alpha \\ \alpha\mu \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} \alpha \\ \alpha\beta \end{smallmatrix} \right\} \left\{ \begin{smallmatrix} \beta \\ \mu\nu \end{smallmatrix} \right\} - \left\{ \begin{smallmatrix} \alpha \\ \beta\mu \end{smallmatrix} \right\} \left\{ \begin{smallmatrix} \beta \\ \nu\alpha \end{smallmatrix} \right\}$, so the Einstein action is given by the Ricci scalar $\mathcal{R} = g^{\mu\nu} \mathcal{R}_{\mu\nu}$ minus the term

$g^{\mu\nu} \left(\partial_\alpha \left\{ \begin{smallmatrix} \alpha \\ \nu\mu \end{smallmatrix} \right\} - \partial_\nu \left\{ \begin{smallmatrix} \alpha \\ \alpha\mu \end{smallmatrix} \right\} \right)$. Hence, neither CGR nor the Einstein action need a GHY boundary term.

- Einstein action is not diffeomorphism invariant, since the connection does not transform like a tensor under coordinate transformations, and the Einstein action picks up boundary terms under such transformations. CGR, on the other hand, is the gauge-fixed version of a perfectly covariant functional, namely the STEGR action.

5.7 The general teleparallel equivalent of general relativity

GTEGR is constructed with a general metric-affine geometry $(\mathcal{M}, g, \Gamma)$ and a single geometric postulate $R_{\mu\nu\rho}^\alpha = 0$.

- We can write the Ricci scalar of the Levi-Civita connection as

$$\begin{aligned} -\mathcal{R}(g) &= \mathbb{T} + \mathbb{Q} + T^{\rho\mu\nu} Q_{\mu\nu\rho} - T^\mu Q_\mu + T^\mu \bar{Q}_\mu + \mathcal{D}_\alpha (Q^\alpha - \bar{Q}^\alpha) + 2T^\alpha \\ &= \mathbb{G} + \mathcal{D}_\alpha (Q^\alpha - \bar{Q}^\alpha) + 2T^\alpha \end{aligned}$$

where we have introduced the scalar $\mathbb{G} = \mathbb{T} + \mathbb{Q} + T^{\rho\mu\nu} Q_{\mu\nu\rho} - T^\mu Q_\mu + T^\mu \bar{Q}_\mu$.

- We then define the action

$$\mathcal{S}_{\text{GTEGR}}[g, \Lambda] = -\frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} \mathbb{G}(g, \Lambda) + \mathcal{S}_{\text{matter}}$$

where $\Lambda \in GL(4, \mathbb{R})$ is the matrix used to parametrize the flat connection. Since the EH action and the action of GTEGR only differ by a total derivative, both actions describe the same theory. Even though GTEGR involves Λ in its definition in addition to g , it enjoys an additional symmetry; $\delta_\Lambda \mathcal{S}_{\text{GTEGR}} = 0$ is satisfied off-shell, which means that the connection is not dynamical. The metric carries physical DOGs while the connection is pure gauge.

- In the absence of non-metricity, $\mathbb{G}$ reduces to $\mathbb{T}$ and TEGR is recovered; similarly, when torsion is absent, $\mathbb{G}$ reduces to $\mathbb{Q}$ and STEGR emerges.

5.8 Non-flat theories

IN TEGR, STEGR, and GTEGR, the crucial step is imposing flatness, immediately implying that the connection has the form $\Gamma = (\Lambda)^{-1} \partial \Lambda$. The Lagrangians defining these three theories are all quadratic in T and Q , and therefore all possess something akin to a kinetic term, $T^2 \sim (\partial \Lambda)^2$ and $Q^2 \sim (\partial \Lambda)^2$. However, if the flatness condition is not imposed, and we allow for torsion and/or non-metricity, we get the actions

$$\begin{aligned} \mathcal{S}_{\text{Einstein-Cartan}}[g, \Gamma] &= \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} (R + \mathbb{T}) + \mathcal{S}_{\text{matter}}[g, \Gamma, \Psi] \\ \mathcal{S}[g, \Gamma] &= \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} (R + \mathbb{Q}) + \mathcal{S}_{\text{matter}}[g, \Gamma, \Psi] \\ \mathcal{S}[g, \Gamma] &= \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{|g|} (R + \mathbb{T} + \mathbb{Q}) + \mathcal{S}_{\text{matter}}[g, \Gamma, \Psi] \end{aligned}$$

which are all deprived of the kinetic term. In all three cases, the connection is a mere auxiliary field which can be integrated out. After integrating out the connection, the resulting actions are not equivalent to GR. Rather, one obtains three modified gravity theories.

5.9 Matter coupling

- In GR, the coupling of the gravitational field to matter fields follows the *minimal coupling principle*: a matter theory formulated in Minkowski space is promoted to a matter theory coupled to the gravitational field $g_{\mu\nu}$ by replacing $\eta_{\mu\nu} \mapsto g_{\mu\nu}$ and $\partial_\mu \mapsto \mathcal{D}_\mu$, provided that the matter fields only couple to $g_{\mu\nu}, g^{\mu\nu}, \sqrt{|g|}$, but not derivatives of the metric.
- Consider TEGR, and naively apply the minimal coupling principle in the form $\eta_{\mu\nu} \mapsto g_{\mu\nu}$ and $\partial_\mu \mapsto \nabla_\mu$, where ∇_μ is the covariant derivative operator w.r.t. the connection. As a specific example, consider the electromagnetic potential A_μ and its associated Maxwell 2-form $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$; according to the minimal coupling principle, the maxwell 2-form becomes $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu - T^\alpha{}_{\mu\nu} A_\alpha$. We immediately see that the minimal coupling principle fails, since the Maxwell action picks up terms proportional to the torsion tensor, thus spoiling the equivalence between TEGR and GR. For fermionic fields, one obtains a similar failure. The Dirac Lagrangian is directly affected by the connection in presence of an axial torsion.
- In STEGR, the minimal coupling principle is preserved even in the presence of non-metricity. In the case of the electromagnetic field A_μ it is straightforward to verify that non-metricity does not contribute to $F_{\mu\nu}$ due to its symmetry and one finds $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu$. For fermions, this property remains unchanged.

6 Modified gravity theories

The actions of TEGR and STEGR are quadratic in the torsion and non-metricity tensors, respectively; one can thus try to construct the most general scalar which is quadratic in the torsion or non-metricity tensor and take this scalar to describe an action functional. In the case of torsion, one finds a three-parameter family of theories described by an action which is quadratic in the torsion tensor; in the case of non-metricity, one finds a five-parameter family of quadratic Lagrangians. Another popular direction is to consider nonlinear extensions of the form $f(\mathbb{T})$, and $f(\mathbb{Q})$, where f is some function which is only subjected to the condition that its first derivative does not vanish.

6.1 Quadratic actions for torsion and non-metricity theories

- Define the trace $T_\mu = T^\alpha{}_{\mu\alpha}$. The most general scalar quadratic in the torsion tensor is a linear combination $\hat{\mathbb{T}} = c_1 T_{\alpha\mu\nu} T^{\alpha\mu\nu} + c_2 T_{\mu\alpha\nu} T^{\alpha\mu\nu} + c_3 T_\mu T^\mu$ where $c_1, c_2, c_3 \in \mathbb{R}$ are arbitrary; $\hat{\mathbb{T}}$ reduces to $\mathbb{T}$ for $c_1 = \frac{1}{4}$, $c_2 = \frac{1}{2}$, $c_3 = -1$. Using this torsion scalar we can write the action functional of *teleparallel gravity*

$$\mathcal{S}_{\text{TG}}[g, \Gamma, \Psi] = - \int_{\mathcal{M}} d^4x \left(\frac{1}{\kappa} \sqrt{|g|} \hat{\mathbb{T}} + \tilde{\Pi}_\alpha{}^{\mu\nu\rho} R^\alpha{}_{\mu\nu\rho} + \tilde{\chi}^\alpha{}_{\mu\nu} Q_\alpha{}^{\mu\nu} \right) + \mathcal{S}_{\text{matter}}[g, \Psi]$$

where the matter fields Ψ are minimally coupled and have a vanishing hypermomentum.

- The metric field equations and connection field equations respectively are

$$\begin{aligned} (\nabla_\alpha + T_\alpha) \hat{S}_{(\mu\nu)}^\alpha + \hat{t}_{\mu\nu} - \frac{1}{2} \hat{\mathbb{T}} g_{\mu\nu} &= \kappa \mathcal{T}_{\mu\nu} \\ (\nabla_\alpha + T_\alpha) \left[\sqrt{|g|} \hat{S}_{[\mu}{}^\alpha{}_{\nu]} \right] &= 0, \end{aligned}$$

where

$$\begin{aligned}\hat{S}_\alpha^{\mu\nu} &= \frac{\partial \hat{\mathbb{T}}}{\partial T^\alpha_{\mu\nu}} = c_1 T_\alpha^{\mu\nu} + c_2 T^{[\mu}{}_\alpha{}^{\nu]} + c_3 \delta_\alpha^{[\mu} T^{\nu]} \\ \hat{t}_{\mu\nu} &= \frac{\partial \hat{\mathbb{T}}}{\partial g^{\mu\nu}} = \frac{1}{2} \hat{S}_{(\mu}{}^{\lambda\kappa} T_{\nu)\lambda\kappa} - T^{\lambda\kappa}{}_{(\mu} \hat{S}_{\lambda\kappa|\nu)}\end{aligned}$$

- The Bianchi identities possess the same form we encountered in TEGR and it is still possible to parametrize the connection by some matrix in $GL(4, \mathbb{R})$. Thus, TEGR and TG look very similar. However, the number of DOFs depends on how one chooses the parameters c, c_2, c_3 .
- Perturbation theory around Minkowski space shows that a self-consistent theory requires $2c_1 + c_2 + c_3 = 0$, which once satisfied leaves one with a 1-parameter family of theories (up to an overall normalization) propagating one additional DOF besides the graviton. Amongst this 1-parameter family of theories, the one which satisfies $2c_1 - c_2$ enjoys an additional symmetry and loses the additional DOF, leaving us with TEGR.
- The most general scalar quadratic in the non-metricity tensor is

$$\hat{\mathbb{Q}} = c_1 Q_{\alpha\mu\nu} Q^{\alpha\mu\nu} + c_2 Q_{\mu\alpha\nu} Q^{\alpha\mu\nu} + c_3 Q_\mu Q^\mu + c_4 \bar{Q}_\mu \bar{Q}^\mu + c_5 Q_\mu \bar{Q}^\mu$$

where the c 's are arbitrary real constants. We can write the functional of *symmetric teleparallel gravity*

$$\mathcal{S}_{\text{STG}}[g, \Gamma, \Psi] = - \int_{\mathcal{M}} d^4x \left(\frac{1}{\kappa} \sqrt{|g|} \hat{\mathbb{Q}} + \tilde{\Pi}_\alpha^{\mu\nu\rho} R^\alpha_{\mu\nu\rho} + \tilde{\chi}_\alpha^{\mu\nu} T^\alpha_{\mu\nu} \right) + \mathcal{S}_{\text{matter}}[g, \Psi]$$

where the matter fields Ψ are assumed to be minimally coupled and having a vanishing hyper-momentum $\mathcal{H}_\alpha^{\mu\nu}$

- The metric field equations and connection field equations respectively are

$$\begin{aligned}\frac{2}{\sqrt{|g|}} \nabla_\alpha \left[\sqrt{|g|} \hat{P}^\alpha_{\mu\nu} \right] + \hat{q}_{\mu\nu} - \frac{1}{2} \hat{\mathbb{Q}} g_{\mu\nu} &= \kappa \mathcal{T}_{\mu\nu} \\ \nabla_\mu \nabla_\nu \left(\sqrt{|g|} \hat{P}^{\mu\nu}{}_\alpha \right) &= 0\end{aligned}$$

where

$$\begin{aligned}\hat{P}^\alpha_{\mu\nu} &:= \frac{1}{2} \frac{\partial \hat{\mathbb{Q}}}{\partial Q^\alpha_{\mu\nu}} = c_1 Q^\alpha_{\mu\nu} + c_2 Q_{(\mu}{}^\alpha{}_{\nu)} + c_3 g_{\mu\nu} Q^\alpha + c_4 \delta^\alpha_{(\mu} \bar{Q}_{\nu)} + \frac{1}{2} c_5 (g_{\mu\nu} \bar{Q}^\alpha + \delta^\alpha_{(\mu} Q_{\nu)}) \\ \hat{q}_{\mu\nu} &:= \frac{\partial \hat{\mathbb{Q}}}{\partial g^{\mu\nu}} = P_{(\mu|\lambda\kappa} Q_{\nu)^{\lambda\kappa}} - 2 P^{\lambda\kappa}{}_{(\mu} Q_{\lambda\kappa|\nu)}\end{aligned}$$

It is still the case that $\hat{\mathbb{Q}} = \hat{P}_{\alpha\mu\nu} Q^{\alpha\mu\nu}$.

- The Bianchi identities, which derive from the diffeomorphism invariance of the action, read $\mathcal{D}_\nu \hat{\mathcal{M}}^\nu{}_\mu + \hat{\mathcal{C}}_\mu = 0$ where $\hat{\mathcal{M}}_{\mu\nu}$ and $\hat{\mathcal{C}}_\alpha$ represent the left hand side of the metric field equations and connection field equations, respectively. Thus, when the metric field equations are satisfied, the connection field equations are identically satisfied as a consequence of the Bianchi identities.

- The number of DOFs in STG also depends on how one chooses the c parameters. Perturbations $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ of Minkowski spacetime lead to a quadratic Lagrangian which is exactly the most general Lagrangian for a spin-2 field. Symmetries have to be imposed in order to remove ghostly DOF's; demanding that the theory is invariant under *transversal diffeomorphisms* $h_{\mu\nu} \mapsto h_{\mu\nu} + 2\partial_{(\mu}\xi_{\nu)}$ where ξ^μ is some vector satisfying $\partial_\mu \xi^\mu = 0$ leads to the condition $2c_1 + c_2 + c_4 = 0$, which is satisfied by the STEGR parameters. In order to recover the two propagating DOFs of a massless spin-2 field, one can further impose *linearized diffeomorphisms*, which are like transversal diffeomorphisms but with the vector field ξ^μ left unrestricted; both conditions taken together imply $c_3 = -c_1, c_4 = -2c_1 - c_2, c_5 = 2c_1$, which matches the defining equations of the sector of STG that STEGR inhabits; STEGR is therefore distinguished through its symmetries and the healthy DOFs it propagates. One could also have imposed instead of linearized diffeomorphisms the *linearized Weyl symmetry* $h_{\mu\nu} \mapsto h_{\mu\nu} + \phi\eta_{\mu\nu}$ for some arbitrary scalar field in addition to the transverse diffeomorphisms, implying $c_3 = -\frac{3}{8}c_1, c_5 = 2c_1$, describing a linearized version of unimodular gravity, which is essentially GR plus the constraint $\sqrt{|g|} = 1$. In unimodular gravity, the cosmological constant emerges from an integration constant.

6.2 Nonlinear extensions

$f(\mathcal{R})$, $f(\mathbb{T})$, $f(\mathbb{Q})$, and $f(\mathbb{G})$ gravity are constructed by simply replacing $\mathcal{R}$, $\mathbb{T}$, $\mathbb{Q}$, and $\mathbb{G}$ in GR, TEGR, STEGR, GTEGR, respectively, by some (generally nonlinear) function f of that scalar. The added freedom in choosing a function f may help in explaining the accelerated expansion of the universe, structure formation, and other phenomena which in the trinity of GR require the introduction of dark energy and dark matter. For example, the most studied and best understood of these theories is $f(\mathcal{R})$ gravity, whose defining action functional is $\mathcal{S}_{f(\mathcal{R})}[g] = \frac{1}{2\kappa} \int_{\mathcal{M}} \sqrt{|g|} f(\mathcal{R}) d^4x$. Given that the original functionals differ by boundary terms, one has to conclude that the resulting non-linear extensions are no longer equivalent to each other; in particular, they each give rise to their own peculiar field equations with their own number of propagating DOFs.

- The equations of $f(\mathcal{R})$ gravity, obtained by following the same route that leads to Einstein's field equations, are

$$f'(\mathcal{R})\mathcal{R}_{\mu\nu} - \frac{1}{2}f(\mathcal{R})g_{\mu\nu} + (g_{\mu\nu}\square - \mathcal{D}_\mu\mathcal{D}_\nu)f'(\mathcal{R}) = \kappa\mathcal{T}_{\mu\nu}$$

where $f'(\mathcal{R}) = \frac{df(\mathcal{R})}{d\mathcal{R}}$ and $\square = g^{\mu\nu}\mathcal{D}_\mu\mathcal{D}_\nu$. In order to avoid the trivial case $f(\mathcal{R}) = \mathcal{R}$, we assume $f''(\mathcal{R}) \neq 0$. The field equations are fourth order non-linear equations for the metric. However, the theory propagates only three healthy DOFs: two DOFs corresponding to a massless graviton and one scalar DOF.

- The metric field equations and connection field equations of $f(\mathbb{T})$ gravity are

$$\begin{aligned} (\nabla_\alpha + T_\alpha) [f'(\mathbb{T})S_{(\mu\nu)}^\alpha] + f'(\mathbb{T})t_{\mu\nu} - \frac{1}{2}f(\mathbb{T})g_{\mu\nu} &= \kappa\mathcal{T}_{\mu\nu} \\ (\nabla_\mu + T_\mu) [f'(\mathbb{T})S_{[\alpha}^\mu{}_{\beta]}] &= 0 \end{aligned}$$

respectively. In practice, it is helpful to rewrite the metric field equations in the form

$$f'(\mathbb{T})G_{\mu\nu} - \frac{1}{2}(f(\mathbb{T}) - f'(\mathbb{T})\mathbb{T}) + f''(\mathbb{T})S_{(\mu\nu)}^\alpha\partial_\alpha\mathbb{T} = \kappa\mathcal{T}_{\mu\nu}$$

It has not yet been determined precisely how many DOFs are propagated by the theory; the number ranges between three and five.

- The metric field equations and connection field equations of $f(\mathbb{Q})$ gravity are

$$-\frac{2}{\sqrt{|g|}}\nabla_\alpha \left[\sqrt{|g|}f'(\mathbb{Q})P^\alpha{}_{\mu\nu} \right] + f'(\mathbb{Q})q_{\mu\nu} - \frac{1}{2}f(\mathbb{Q})g_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu}$$

$$\nabla_\mu \nabla_\nu \left(\sqrt{|g|}f'(\mathbb{Q})P^{\mu\nu}{}_\alpha \right) = 0$$

respectively. It is possible to rewrite the metric field equations as

$$f'(\mathbb{Q})G_{\mu\nu} - \frac{1}{2}(f(\mathbb{Q}) - f'(\mathbb{Q})\mathbb{Q})g_{\mu\nu} + 2f''(\mathbb{Q})P^\alpha{}_{\mu\nu}\partial_\alpha \mathbb{Q} = \kappa \mathcal{T}_{\mu\nu}$$

- The Bianchi identities are found by following the same reasoning as in GR/TEGR/STEGR; one finds $\mathcal{D}_\mu \mathcal{M}^\mu{}_\nu + \mathcal{C}_\nu = 0$, where $\mathcal{M}_{\mu\nu}, \mathcal{C}_\alpha$ are the left hand sides of the above metric/connection field equations, respectively. Indeed, when the metric field equations are satisfied, then the connection field equations are also satisfied due to the Bianchi identities.
- $f(\mathbb{Q})$ gravity can contain the GR solutions as special cases. If we impose the condition $\mathbb{Q} = \mathbb{Q}_0 = \text{const}$, and if we assume that we can actually satisfy this condition, it follows that the metric field equations take the form $G_{\mu\nu} - \frac{1}{2}\frac{f(\mathbb{Q}_0)-f'(\mathbb{Q}_0)\mathbb{Q}_0}{f'(\mathbb{Q}_0)}g_{\mu\nu} = \frac{\kappa}{f'(\mathbb{Q}_0)}\mathcal{T}_{\mu\nu}$ which can be read as Einstein's field equations with an effective cosmological constant $-\frac{1}{2}\frac{f(\mathbb{Q}_0)-f'(\mathbb{Q}_0)\mathbb{Q}_0}{f'(\mathbb{Q}_0)}$ and a rescaled energy-momentum tensor.
- The metric field equations and connection field equations of $f(\mathbb{G})$ gravity are

$$f'(\mathbb{G})G_{\mu\nu} - \frac{1}{2}(f(\mathbb{G}) - f'(\mathbb{G})\mathbb{G})g_{\mu\nu} + \mathcal{D}_\mu f'(\mathbb{G})M^\sigma{}_{\nu\sigma}$$

$$+ f''(\mathbb{G}) \left(M^{[\rho\sigma]}{}_\sigma g_{\mu\nu} - M^\rho{}_{(\mu\nu)} \right) \partial_\rho \mathbb{G} = \kappa \mathcal{T}_{\mu\nu}$$

$$\nabla_\rho (f'(\mathbb{G})Z_\mu{}^{\nu\rho}) - f'(\mathbb{G})M^\lambda{}_{\rho\lambda}Z_\mu{}^{\nu\rho} = 0$$

respectively. Imposing either vanishing non-metricity or torsion, which should be read as partial gauge-fixing conditions for the connection, one recovers the $f(\mathbb{T})$ and $f(\mathbb{Q})$ field equations respectively. Imposing the condition $\mathbb{G} = \mathbb{G}_0 = \text{const}$ reduces the field equations to $G_{\mu\nu} - \frac{1}{2}\frac{f(\mathbb{G}_0)-f'(\mathbb{G}_0)\mathbb{G}_0}{f'(\mathbb{G}_0)}g_{\mu\nu} = \frac{\kappa}{f'(\mathbb{G}_0)}\mathcal{T}_{\mu\nu}$ which are again Einstein's field equations with an effective cosmological constant and rescaled energy tensor.

7 Dynamics of the homogeneous universe

- The *cosmological principle* is that we should not assume that we are the center of the universe.
- The universe should appear isotropic to any free-falling observer throughout the universe. If the universe is isotropic around all points, then it is also homogeneous. To a first approximation, we will therefore treat the universe as perfectly homogeneous and isotropic.
- Spatial homogeneity and isotropy mean that the universe can be represented by a time-ordered sequence of three-dimensional spatial slices, each of which is homogeneous and isotropic. Such

spaces are the maximally symmetric 3-spaces. Homogeneous and isotropic 3-spaces have constant curvature, which must be zero, positive, or negative. A 3-space with constant positive curvature can be represented as a 3-sphere S^3 embedded in four-dimensional Euclidean space E^4 , and a 3-space with constant negative curvature can be represented as a hyperboloid H^3 embedded in four-dimensional Lorentzian space $\mathbb{R}^{1,3}$.

- After a rescaling of coordinates $x^i \rightarrow ax^i$, where a is the radius of the 3-sphere in the spherical case and a^2 is an arbitrary constant in the arbitrary constant, the line element of a general maximally symmetric 3-space can be written

$$d\ell^2 = a^2 \gamma_{ij} dx^i dx^j$$

$$\text{where } \gamma_{ij} = \delta_{ij} + k \frac{x_i x_j}{1 - k(x_k x^k)}, \text{ for } k = \begin{cases} 0 & \text{Euclidean} \\ +1 & \text{spherical} \\ -1 & \text{hyperbolic} \end{cases}$$

Notice we must take $a^2 > 0$ in order to have $d\ell^2$ positive at $x^i = 0$, and hence everywhere.

- The form of the spatial metric γ_{ij} depends on the choice of coordinates. In spherical polar coordinates, the metric becomes diagonal:

$$d\ell^2 = a^2 \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$. We may also define a radial coordinate $d\chi = dr/\sqrt{1 - kr^2}$ so that

$$d\ell^2 = a^2 [d\chi^2 + S_k^2(\chi) d\Omega^2] \text{ where } S_k(\chi) = \begin{cases} \sinh \chi & k = -1 \\ \chi & k = 0 \\ \sin \chi & k = +1 \end{cases}$$

- The *(Friedmann-)Robertson-Walker/FRW metric* for an expanding universe is obtained by including $d\ell^2$ as defined above into the spacetime line element and let the parameter a be an arbitrary function of time:

$$ds^2 = dt^2 - a^2(t) \gamma_{ij} dx^i dx^j$$

- Homogeneity and isotropy therefore reduce the ten independent coordinates of the spacetime metric into a *scale factor* $a(t)$ and a curvature parameter k .
- The coordinates x^i are called *comoving coordinates*, and the coordinates $x_{\text{phys}}^i = a(t)x^i$ are the *physical coordinates*. The physical velocity of an object is $v_{\text{phys}}^i = \frac{dx_{\text{phys}}^i}{dt} = a(t) \frac{dx^i}{dt} + \frac{da}{dt} x^i = v_{\text{pec}}^i + H x_{\text{phys}}^i$ where we have defined the *peculiar velocity* $v_{\text{pec}}^i = a(t) \dot{x}^i$ and the *Hubble parameter* $H = \frac{\dot{a}}{a}$. The $H x_{\text{phys}}^i$ term is called the *Hubble flow*. The peculiar velocity of an object is the velocity measured by a comoving observer, i.e. an observer who follows the Hubble flow.
- The FRW metric in polar coordinates reads

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

The line element has a rescaling symmetry $a \rightarrow \lambda a$, $r \rightarrow r/\lambda$, $k \rightarrow \lambda^2 k$. We then set $a_0 = a(t_0) = 1$.

- We can also write $ds^2 = dt^2 - a^2(t) [d\chi^2 + S_k^2(\chi)d\Omega^2]$; defining *conformal time* $d\tau = \frac{dt}{a(t)}$, this becomes $ds^2 = a^2(\tau)[d\tau^2 - (d\chi^2 + S_k^2(\chi)d\Omega^2)]$, which is a static Minkowski metric multiplied by a time-dependent conformal factor $a(\tau)$. Since light travels along null geodesics, the propagation of light in FRW is the same as in Minkowski space if we first transform to conformal time. Along the path, the change in conformal time equals the change in comoving distance.
- In this metric, the redshift for a photon emitted by a distant galaxy at time t_1 and observed on earth today is $z = \frac{a(t_0)}{a(t_1)} - 1$, where we recall that $a(t_0)$ is often defined to be 1.
- For nearby sources, we may expand $a(t_1)$ in a power series $a(t_1) = a(t_0)[1 + (t_1 - t_0)H_0 + \dots]$ where H_0 is the *Hubble constant* $H_0 = \frac{\dot{a}(t_0)}{a(t_0)}$. For close objects, $t_0 - t_1$ is simply the physical distance d (in units with $c = 1$). This gives us *Hubble's law* $z \approx H_0 d$. The slope in a redshift-distance diagram therefore measures the current expansion rate of the universe H_0 .
- Write the FRW metric $ds^2 = dt^2 - a^2(t)[d\chi^2 + S_k^2(\chi)d\Omega^2]$ where we have used the rescaling symmetry $a \rightarrow \lambda a$, $\chi \rightarrow \chi/\lambda$, $S_k^2 \rightarrow S_k^2/\lambda$ after choosing the scale factor to be dimensionless ($a(t_0) = 1$) to rewrite

$$S_k(\chi) = \begin{cases} R_0 \sinh(\chi/R_0) & k = -1 \\ \chi & k = 0 \\ R_0 \sin(\chi/R_0) & k = +1 \end{cases}$$

We use this to define the *metric distance* $d_m = S_k(\chi)$. In a flat universe, the metric distance equals to *comoving distance* χ ; the comoving distance between us and a galaxy at redshift z is $\chi(z) = \int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^z \frac{dz}{H(z)}$, where the redshift evolution of the Hubble parameter depends on the matter content of the universe.

- Recall that absolute luminosity L is energy emitted per second and flux F is energy per second per receiving area. Type IA supernovae are called standard candles because they are believed to be objects of known absolute luminosity. The observed flux from a supernova explosion can be used to infer its (luminosity) distance. For a source at a fixed comoving distance χ in a static Euclidean space, $F = \frac{L}{4\pi\chi^2}$. In an FRW spacetime, this result is modified; at the time t_0 that the light reaches earth, the proper area of a sphere drawn around the supernova passing through Earth is $4\pi d_m^2$, so the fraction of light received in a telescope of aperture A is $A/4\pi d_m^2$. Moreover, the rate of arrival of photons is lower than the rate at which they are emitted, and the energy of the photons when they are received is less than the energy with which they are emitted. Hence in FRW spacetime

$$F = \frac{L}{4\pi d_m^2(1+z)^2} = \frac{L}{4\pi d_L^2}$$

where we have defined the *luminosity distance* $d_L = d_m(1+z)$.

- Standard rulers are objects of known physical size D . Assuming that the object is at a comoving distance χ and the photons which we observe today were emitted at time t_1 , one could naively use for the distance between the observer and the emitter the *angular diameter distance* $d_A = \frac{D}{\delta\theta}$, where $\delta\theta$ is its angular size; the FRW metric implies that $D = a(t_1)S_k(\chi)\delta\theta = \frac{d_m}{1+z}\delta\theta$. We see that $d_A = \frac{d_L}{(1+z)^2}$.

- Recall the covariant conservation equation $\nabla_\mu T^\mu_\nu = 0$. The evolution of energy density is determined by the $\nu = 0$ equation. In an isotropic universe $T^i_0 = 0$, so this reduces to $\dot{\rho} + \Gamma^\mu_{\mu 0}\rho - \Gamma^\lambda_{\mu 0}T^\mu_\lambda = 0$, which in the FRW metric reads $\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + P) = 0$.
- The universe is filled with many different things, which we can classify by their contribution to pressure.
 - *Matter* will be used to refer to all forms of matter for which pressure is much smaller than the energy density; in this case $\rho \propto a^{-3}$. This dilution of the energy density reflects the expansion of the volume $V \propto a^3$.
 - * Cosmologists refer to ordinary matter (nuclei and electrons) as *baryons*. Disgraceful.
 - * Most of the matter, however, is *dark matter*. We haven't figured out that part yet.
 - *Radiation* is anything for which pressure is about a third of the energy density, $P = \frac{1}{3}\rho$. This is the case for a gas of relativistic particles, for which the energy density is dominated by the kinetic energy. In this case $\rho \propto a^{-4}$. The dilution now includes the redshifting of energy $E \propto a^{-1}$.
 - * *Photons* dominated the early universe. We detect them in those photons in the form of the CMB.
 - * *Neutrinos* behaved like radiation for most of the history of the universe, but recently their small masses have become relevant and they're starting to behave like matter.
 - * *Gravitons* might exist.
 - The universe today seems to be dominated by a mysterious negative pressure component $P = -\rho$, which we call *dark energy*. In particular, the energy density of dark energy is constant; since the energy density doesn't dilute, energy has to be created as the universe expands. Note that this does not violate the conservation of energy in the gravitational context, since the proper conservation equation is the continuity equation above. Quantum field theory predicts this effect in the form of *vacuum energy*; the ground state energy of the vacuum corresponds to the stress-energy tensor $T^{\text{vac}} = \rho_{\text{vac}}g_{\mu\nu}$. Unfortunately, the predicted size of ρ_{vac} is greater than the observed ρ by a factor of 10^{120} .
- We note before moving on that the left-hand side of the Einstein equation $G_{\mu\nu} = 8\pi GT_{\mu\nu}$ isn't uniquely defined, and that we can add the term $-\Lambda g_{\mu\nu}$ for some *cosmological constant* Λ without violating $\nabla^\mu T_{\mu\nu} = 0$. Modern practice is to move this term to the right hand side and treat it as a contribution to the stress-energy tensor of the form $T_{\mu\nu}^{(\Lambda)} = \frac{\Lambda}{8\pi G}g_{\mu\nu} = \rho_\Lambda g_{\mu\nu}$. This is the same form as the stress-energy tensor from vacuum energy.
- In the FRW metric, the non-vanishing components of the Ricci tensor are

$$R_{00} = -3\frac{\ddot{a}}{a},$$

$$R_{ij} = \left[\frac{\ddot{a}}{a} + 2\left(\frac{\dot{a}}{a}\right)^2 + 2\frac{k}{a^2} \right] g_{ij}$$

and the Ricci scalar is

$$R = -6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} \right]$$

The nonzero components of the Einstein tensor are

$$G_0^0 = 3 \left[\left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right]$$

$$G_j^i = \left[2 \frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{k}{a^2} \right] \delta_j^i$$

- Recall that the stress-energy tensor for a perfect fluid in a rest frame is $T_\nu^\mu = \text{diag}(\rho, -P, -P, -P)$ (where the signs on the P simply come from our change in metric signature). Using this, we get the *Friedmann equations*

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3P)$$

where ρ and P are understood as the sum of all contributions to the energy density and pressure in the universe. The first Friedmann equation can also be written in terms of the Hubble parameter as

$$H^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2}$$

- Use superscripts 0 to denote quantities evaluated today at $t = t_0$. A flat universe $k = 0$ corresponds to the *critical density* today $\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G}$, which we use to define dimensionless density parameters $\Omega_{I,0} = \frac{\rho_{I,0}}{\rho_{\text{crit},0}}$.
- We write ρ_r for contribution from radiation (ρ_γ for photons, ρ_ν for neutrinos), ρ_m for matter (ρ_c for cold dark matter, ρ_b for baryons), and ρ_Γ for vacuum energy.
- The first Friedmann equation can then be written

$$H^2(a) = H_0^2 \left[\Omega_{r,0} \left(\frac{a_0}{a} \right)^4 + \Omega_{m,0} \left(\frac{a_0}{a} \right)^3 + \Omega_{k,0} \left(\frac{a_0}{a} \right)^2 + \Omega_{\Gamma,0} \right]$$

where we have defined a curvature density parameter $\Omega_{k,0} = -k/(a_0 H_0)^2$. In literature, the subscript 0 is normally dropped (so it is assumed that we speak of today unless otherwise stated). We will also use $a_0 = 1$.

- Observations show that $|\Omega_k| \leq 0.01$, $\Omega_r = 9.4 \times 10^{-5}$, $\Omega_m = 0.32$, $\Omega_\Lambda = 0.68$. The equation of state of dark energy appears to be that of a cosmological constant $w_\Lambda \approx 1$. Matter splits into $\Omega_b = 0.05$ and $\Omega_c = 0.27$. Since curvature makes up less than 1% of the cosmic energy budget we set $\Omega_k = 0$ in what follows.
- The different scalings of radiation (a^{-4}), matter (a^{-3}), and vacuum energy (a^0) imply that for most of its history the universe was dominated by a single component (first radiation, then matter, then vacuum energy). We parametrize this component by its equation of state w_I . For a flat, single-component universe, the Friedmann equation reduces to

$$\frac{\dot{a}}{a} = H_0 \sqrt{\Omega_I} a^{-\frac{3}{2}(1+w_I)}$$

REMEMBER TO LATEX TABLE 1.1 BAUMANN PG 31

- At $a_{eq} = \Omega_r/\Omega_m \approx 3 \times 10^{-4}$, we had a two-component universe. In a flat universe filled with a mixture of matter and radiation, the Friedmann equations are

$$(a')^2 = \frac{8\pi G}{3} \rho a^4$$

$$a'' = \frac{4\pi G}{3} (\rho - 3P) a^3$$

where primes denote derivatives w.r.t. conformal time and $\rho = \rho_m + \rho_r = \frac{\rho_{eq}}{2} \left[\left(\frac{a_{eq}}{a} \right)^3 + \left(\frac{a_{eq}}{a} \right)^4 \right]$. Since $\rho_r - 3P_r = 0$ and $\rho_m a^3 = \frac{1}{2} \rho_{eq} a_{eq}^3$, it follows that $a'' = \frac{2\pi G}{3} \rho_{eq} a_{eq}^3$. Imposing $a(\tau = 0) = 0$ fixes one of the constants of integration, so the differential equation has the solution

$$a(\tau) = a_{eq} \left[\left(\frac{\tau}{\tau_*} \right)^2 + 2 \left(\frac{\tau}{\tau_*} \right) \right], \text{ where } \tau_* = \left(\frac{\pi G}{3} \rho_{eq} a_{eq}^2 \right)^{-1/2} = \frac{\tau_{eq}}{\sqrt{2} - 1}$$

For $\tau \ll \tau_{eq}$, we recover the radiation-dominated limit $a \propto \tau$, while for $\tau \gg \tau_{eq}$ we get the matter-dominated limit $a \propto \tau^2$.

8 Cosmological inflation

- Standard cosmology predicts that the early universe was made of many causally disconnected regions of space. The *horizon problem* is to explain that why these apparently disjoint patches of space have very nearly the same densities and temperatures.
- Define the reduced Planck mass $M_{pl} = \sqrt{\frac{\hbar c}{8\pi G}}$ so that the Friedmann equation is $H^2 = \rho/(3M_{pl}^2)$.
- In an isotropic spacetime we can always define the coordinate system so that light travels purely in the radial direction. The evolution is then determined by a two-dimensional line element $ds^2 = a^2(\tau)[d\tau^2 - d\chi^2]$ (where we have again used the reparametrization $d\chi = dr/\sqrt{1 - kr^2}$ to deal with curvature); photons travel along the null geodesics with paths $\Delta\chi(\tau) = \pm\Delta\tau$, where the plus sign corresponds to outgoing photons and the minus to incoming photons.
- The maximal comoving distance that light can travel between two times $\tau_1, \tau_2 > \tau_1$ is simply $\tau_2 - \tau_1$; if the Big Bang starts with a singularity at $t_i = 0$ (note the Big Bang singularity is a moment in time but not space), then the greatest comoving distance from which an observer at time t will be able to receive signals travelling at the speed of light is given by $\chi_{ph}(\tau) = \tau - \tau_i = \int_{\tau_i}^{\tau} \frac{dt}{a(t)}$. This is the (*comoving*) *particle horizon*. The size of the particle horizon at τ may be visualized by the intersection of the past light cone of an observer p with the spacelike hypersurface $\tau = \tau_i$. Causal influences have to come from within this region. Only comoving particles whose worldlines intersect that past light cone of p can send a signal to an observer at p ; the boundary of the region containing such worldlines is the particle horizon at p .
- In comoving coordinates, the greatest distance from which an observer at time t_f will receive signals emitted at any time later than t is given by $\chi_{eh}(\tau) = \tau_f - \tau = \int_{\tau}^{\tau_f} \frac{dt}{a(t)}$. This is called the (*comoving*) *event horizon*. τ_f denotes the final moment of (conformal) time. This may be finite even if physical time is infinite; whether or not this is the case depends on the form of $a(t)$. τ_f is finite for our universe if dark energy is really a cosmological constant.

- We can write $\chi_{ph}(\tau) = \int_{a_i}^a \frac{da}{a\dot{a}} = \int_{\ln a_i}^{\ln a} (aH)^{-1} d \ln a$ where $a_i = 0$ corresponds to the Big Bang singularity. We call $(aH)^{-1}$ the *comoving Hubble radius*. For a universe dominated by a fluid with constant equation of state $w = P/\rho$, we have $(aH)^{-1} = H_0^{-1} a^{\frac{1}{2}(1+3w)}$. All familiar matter sources satisfy the strong energy condition $1 + 3w > 0$, so we used to assume that the comoving Hubble radius increases as the universe expands, in which case the integral is dominated by the upper limit and receives vanishing contributions from early times, which is explicit in the example of a perfect fluid, in which case $\chi_{ph}(a) = \frac{2H_0^{-1}}{(1+3w)} \left[a^{\frac{1}{2}(1+3w)} - a_i^{\frac{1}{2}(1+3w)} \right] = \tau - \tau_i$. Defining $\tau_i = \frac{2H_0^{-1}}{(1+3w)} a_i^{\frac{1}{2}(1+3w)} \xrightarrow{a_i \rightarrow 0, w > -\frac{1}{3}} 0$, the comoving horizon is finite, $\chi_{ph}(t) = \frac{2}{(1+3w)} (aH)^{-1}$. So in standard cosmology, $\chi_{ph} \sim (aH)^{-1}$.
- Conjecture a phase of decreasing Hubble radius in the universe. This requires a strong-energy-condition-violating fluid. The integral above is dominated by the lower limit; the Big Bang singularity is pushed to negative conformal time, $\tau_i \rightarrow -\infty$. The widely separated points in the CMB now have enough time to intersect before the time τ_i . In inflationary cosmology, $\tau = 0$ isn't the initial singularity, but instead becomes only a transition point between inflation and the standard Big Bang evolution; there is time both before and after $\tau = 0$.
- From the relation $\frac{d}{dt}(aH)^{-1} = \frac{d}{dt}(\dot{a})^{-1} = -\frac{\ddot{a}}{(\dot{a})^2}$, we see that a shrinking comoving Hubble radius implies accelerated expansion $\ddot{a} > 0$. As such, inflation is often defined as a period of acceleration.
- Alternatively, we may write $\frac{d}{dt}(aH)^{-1} = -\frac{\dot{a}H + a\dot{H}}{(aH)^2} = -\frac{1}{a}(1 - \varepsilon)$ where $\varepsilon = -\frac{\dot{H}}{H^2}$; the shrinking Hubble sphere therefore also corresponds to $\varepsilon < 1$.
- For perfect inflation $\varepsilon = 0$, the spacetime becomes *de Sitter space* $ds^2 = dt^2 - e^{2Ht} dx^2$ where $H = \partial_t \ln a = \text{const}$. Inflation has to end, so it shouldn't correspond to perfect de Sitter space; for small but finite $\varepsilon \neq 0$, this line element is still a good approximation to the inflationary background, so we sometimes refer to inflation as a quasi-de Sitter period.
- For a perfect fluid with pressure P and density ρ , the Friedmann equation $H^2 = \rho/(3M_{pl}^2)$ and continuity together imply $\varepsilon = \frac{3}{2} \left(1 + \frac{P}{\rho} \right) < 1 \iff w = \frac{P}{\rho} < -\frac{1}{3}$, i.e. that inflation requires negative pressure or a violation of the SEC. Moreover, $\left| \frac{d \ln \rho}{d \ln a} \right| = 2\varepsilon < 1$, so for small ε the energy density is nearly constant.
- Therefore, a given FRW spacetime with time-dependent $H(t)$ corresponds to cosmic acceleration iff $\varepsilon = -\frac{\dot{H}}{H^2} = -\frac{d \ln H}{dN} < 1$, where $dN = H dt$ measures the number of e -folds N of inflationary expansion. Solving the horizon problem requires N to be at least 40 to 60 e -folds, requiring ε to be small for long enough; this condition is measured by $\eta = \frac{d \ln \varepsilon}{dN} = \frac{\ddot{H}}{H\dot{H}}$, since for $|\eta| < 1$, the fractional change of ε per Hubble time is small and inflation persists.
- As a toy model for inflation consider the scalar field $\phi(t, \vec{x})$ which we call the *inflaton*. Consistency with the symmetries of FRW spacetime requires that the background value of the inflaton only depends on time. From the time component $T_0^0 = \rho_\phi$ of the stress-energy tensor, we infer that $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$, and from the space-space component $T_j^i = -P_\phi \delta_j^i$, we find that the pressure is $P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi)$. Therefore, a field configuration leads to inflation, $P_\phi < -\frac{1}{3}\rho_\phi$, if potential energy dominates over the kinetic energy.

- After substituting ρ_ϕ into the first Friedmann equation, we get

$$H^2 = \frac{1}{3M_{pl}^2} \left[\frac{1}{2}\dot{\phi}^2 + V \right]$$

And from the second Friedmann equation

$$\dot{H} = -\frac{1}{2} \frac{\dot{\phi}^2}{M_{pl}^2}$$

Combining this with the equation obtained from taking the time derivative of the first equation, we get the Klein-Gordon equation

$$\ddot{\phi} + 3H\dot{\phi} + V' = 0$$

The potential acts like a force, while the expansion of the universe adds friction.

- From the above we find $\varepsilon = \frac{\frac{1}{2}\dot{\phi}^2}{M_{pl}^2 H^2}$. Inflation therefore occurs if the kinetic energy makes only a small contribution to the total energy $\rho_\phi = 3M_{pl}^2 H^2$. This situation is called *slow-roll inflation*. For this condition the acceleration of the scalar field has to be small. Define $\delta = -\frac{\ddot{\phi}}{H\dot{\phi}}$. Then substituting things gives $\eta = 2(\varepsilon - \delta)$. Hence, $\{\varepsilon, |\delta|\} \ll 1 \implies \{\varepsilon, |\eta|\} \ll 1$.
- Using these conditions to simplify the equations of motion is called the *slow-roll approximation*. The condition $\varepsilon \ll 1$ implies $\frac{1}{2}\dot{\phi}^2 \ll V$, leading to the simplification of the Friedmann equation,

$$H^2 \approx \frac{V}{3M_{pl}^2}$$

In the slow-roll approximation, the Hubble expansion is determined completely by the potential energy; $|\delta| \ll 1$ simplifies the Klein-Gordon equation to

$$3H\dot{\phi} \approx -V'$$

From these, $\delta + \varepsilon \approx M_{pl}^2 \frac{V''}{V}$, hence we can assess whether a given potential $V(\phi)$ leads to slow-roll inflation is to compute the *potential slow-roll parameters*

$$\epsilon_v = \frac{M_{pl}^2}{2} \left(\frac{V'}{V} \right)^2, \quad |\eta_v| = M_{pl}^2 \frac{|V''|}{V}$$

Successful slow-roll inflation occurs when these parameters are small. In contrast, η, ε are called *Hubble slow-roll parameters*; during slow roll, $\epsilon_v \approx \varepsilon$ and $\eta_v \approx 2\varepsilon - \frac{1}{2}\eta$.

- The total number of e -folds of accelerated expansion are $N_{tot} = \int_{a_I}^{a_E} d \ln a = \int_{t_I}^{t_E} H(t) dt$, where t_I, t_E are defined as the times when $\varepsilon(t_I) = \varepsilon(t_E) = 1$. In the slow-roll regime, $H dt = \frac{1}{\sqrt{2\varepsilon}} \frac{|d\phi|}{M_{pl}} \approx \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}}$, so $N_{tot} = \int_{\phi_I}^{\phi_E} \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}}$ where ϕ_I, ϕ_E are the boundaries of the interval where $\epsilon_v < 1$. The largest scales observed in the CMB are produced about 60 e -folds before the end of inflation; $N_{cmb} = \int_{\phi_{cmb}}^{\phi_E} \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}} \approx 60$. A successful solution to the horizon problem requires $N_{tot} > N_{cmb}$.
- During inflation, most of the energy density in the universe is in the form of the inflaton potential; inflation ends when the potential steepens and the inflaton field picks up kinetic energy. The energy in the inflaton sector is then transferred to the Standard Model particles. This process is called *reheating* and begins with the *Hot Big Bang*.

- After inflation, ϕ begins to oscillate at the bottom potential. Assuming $V(\phi) \approx \frac{1}{2}m^2\phi^2$ near the minimum of $V(\phi)$ where the amplitude of ϕ is small, the inflaton is still homogeneous, so its EOM is $\ddot{\phi} + 3H\dot{\phi} = -m^2\phi$. The expansion time scale becomes much longer than the oscillation period, $H^{-1} \gg m^{-1}$. We can then neglect the friction term and the field undergoes oscillations with frequency m . Writing continuity as $\dot{\rho}_\phi + 3H\rho_\phi = -3HP_\rho = -\frac{3}{2}H(m^2\phi^2 - \dot{\phi}^2)$, the r.h.s. averages to zero over one oscillation period, so the oscillating field therefore behaves like pressureless matter with $\rho_\phi \propto a^{-3}$. The fall in energy density is reflected in a decrease of the oscillation amplitude.
- The inflaton has to couple to SM fields, so the energy stored in it will then be transferred to ordinary particles. If the decay is slow, in which case the inflaton can only decay into fermions, the inflaton energy density follows $\dot{\rho}_\phi + 3H\rho_\phi = -\Gamma_\phi\rho_\phi$, where Γ_ϕ parameterizes inflaton decay rate. If the inflaton can decay into bosons, the decay may be very rapid, involving *parametric resonance* sourced by Bose condensation events. This kind of rapid decay is called *preheating*, since the bosons are far from thermal equilibrium.
- The particles produced by inflaton decay will interact and the resulting particle soup will eventually reach thermal equilibrium with some temperature T_{rh} . Reheating temperature is determined by the energy density ρ_{rh} at the end of the reheating epoch. $\rho_{rh} < \rho_{\phi,E}$ where $\rho_{\phi,E}$ is the inflaton energy density at the end of inflation; if reheating takes a long time, $\rho_{rh} \ll \rho_{\phi,E}$. Though it can be quite involved, the evolution of the gas particles into a thermal state is assumed to happen eventually; some particles that never reach thermal equilibrium (e.g. gravitinos) might exist, since their interactions are too weak. As long as the momenta of the particles are much higher than their masses, the energy density of the universe behaves like radiations regardless of the momentum space distribution. After thermalisation of at least the baryons, photons, and neutrinos is complete, the standard Hot Big Bang era begins.

9 $f(\mathbb{Q})$ gravity

9.1 Cosmology in $f(\mathbb{Q})$

Given that the coincident gauge can always be used in STG, the we the simplest thing to do is to use this gauge plus a fixed background metric.

- Begin with the ansatz $\Gamma_{\mu\nu}^\alpha = 0$, $g_{\mu\nu} = \text{diag}(-N(t)^2, a(t)^2, a(t)^2, a(t)^2)$ where $N(t), a(t)$ are the usual lapse function and scale factor of FLRW spacetime.
- The non-metricity scalar for this ansatz is simply $\mathbb{Q} = 6\frac{H^2}{N^2}$ where $H = \frac{\dot{a}}{a}$ is the usual Hubble function.
- The symmetry-reduced action

$$\mathcal{S}[N, a] = -\frac{1}{2\kappa} \int_{\mathcal{M}} dt d^3\vec{x} a(t)^3 N(t) f(\mathbb{Q})$$

which is the $f(\mathbb{Q})$ action evaluated on the FLRW metric and in the coincident gauge, has a residual time-reparametrization invariance; exploiting this, we can fix the lapse function to

unity and obtain the cosmological equations

$$6f'H^2 - \frac{1}{2}f = \rho$$

$$(12H^2f'' + f')\dot{H} = -\frac{1}{2}(\rho + p)$$

where ρ, p denote density, pressure, respectively, and where $f' = \frac{df}{d\mathbb{Q}}$, $f'' = \frac{d^2f}{d\mathbb{Q}^2}$. Standard matter fields also satisfy continuity $\dot{\rho} = -3H(\rho + p)$.

- If we demand that f satisfy $6f'H^2 - \frac{1}{2}f = \frac{1}{2\kappa}\mathbb{Q}$ with $\kappa = 8\pi G$, this gives us the same background evolution as GR, but the evolution of perturbations is subjected to modifications; this condition is a differential equation solved by $f(\mathbb{Q}) = \frac{1}{\kappa}(\mathbb{Q} + M\sqrt{\mathbb{Q}})$, where M is an integration constant.
- Another interesting ansatz for studying $f(\mathbb{Q})$ cosmology is $f(\mathbb{Q}) = \frac{1}{\kappa}\left[\mathbb{Q} - 6\lambda M^2\left(\frac{\mathbb{Q}}{6M^2}\right)^\alpha\right]$ where λ, α are dimensionless parameters; the modified Friedmann equation for this ansatz reads

$$H^2 \left[1 + (1 - 2\alpha)\lambda \left(\frac{H^2}{M^2} \right)^{\alpha-1} \right] = \frac{\kappa}{3}\rho$$

The previous f is a special case for $\alpha = \frac{1}{2}$. $\alpha = 1$ recovers STEGR. For $\alpha < 1$ the corrections to the GR evolution become important at low curvature, while for $\alpha > 1$ corrections become relevant in the high curvature regime. In other words, $\alpha > 1$ theories play a role in the early universe and $\alpha < 1$ theories provide corrections to late-time cosmology.

- Because in $f(\mathbb{Q})$ gravity the connection field equations are no longer trivial identities, $f(\mathbb{Q})$ can be expected to propagate more than the two DOFs of GR; when working in the coincident gauge and using the FLRW metric as an ansatz, interesting cosmological models can emerge, but we become largely oblivious to additional DOFs because of these overly restrictive choices. Perturbation theory around FLRW while still in the coincident gauge leads to the insight that $f(\mathbb{Q})$ gravity propagates at least two additional DOFs. One can also abandon the coincident gauge and instead work with a metric and connection both compatible with homogeneity and isotropy.

9.2 Black holes in $f(\mathbb{Q})$

The simplest possible strategy is to fix a coincident gauge and choose a metric ansatz which is stationary and spherically symmetric. However, the field equations for $f(\mathbb{Q})$ are inconsistent with this metric ansatz in the coincident gauge, except in the special case $f'' = 0$, i.e. the inconsistency disappears precisely when we are in the regime of the theory equivalent to GR. To overcome this, no special metric nor any special gauge for the connection can be assumed. We instead find the most general metric and connection which are stationary and spherically symmetric; using these, we implement the postulates of vanishing curvature and vanishing torsion, and take the metric and connection which satisfy all symmetries and geometric postulates and plug them into the $f(\mathbb{Q})$ field equations.